Cold Days, Warm Homes: The Impact of Winter Weather on Remote Work *

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Abstract

This study examines how winter weather conditions affect remote work patterns across U.S. metropolitan statistical areas. Combining individual-level telework data from the Current Population Survey with daily weather records from the National Oceanic and Atmospheric Administration, I employ a two-way fixed effects model and a dose-response extension of a staggered differences-in-differences framework to estimate the causal impact of winter weather on working hours and their allocation across work locations, and to trace the dynamic dose-response relationship between snowfall and telework hours. The empirical results demonstrate that adverse winter conditions increase telework hours, with the clearest and most consistent effect arising from snowfall, and that this response is markedly stronger in occupations characterized by high telework feasibility. Furthermore, while snowfall reduces on-site work, telework partially compensates for the lost hours, mitigating the overall decline in total hours worked. The estimated dose-response curve indicates a positive relationship between snowfall intensity and telework hours. These findings suggest that spatial reallocation of work serves as a partial adaptation mechanism against weather-induced labor disruptions, with the effects concentrated among workers in telework-feasible occupations, offering implications for climate adaptation policy and infrastructure planning.

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1 Introduction

Inclement winter weather can severely disrupt economic activity by paralyzing the movement of people and goods. In December 2022, for instance, Winter Storm Elliott dropped more than four feet of snow on the Buffalo metropolitan area over five days, led airlines to cancel over 18,000 flights nationwide, and prompted local authorities to impose a driving ban that lasted more than five days, halting commuting and on-site work across the region. Classified by NOAA as a billion-dollar weather and climate disaster, the storm left more than a million customers without power and placed over 200 million people under winter weather advisories (National Centers for Environmental Information, 2025). Such severe winter events are not isolated occurrences. Although global warming has made midlatitude cold extremes fewer and milder in recent decades (Blackport et al., 2024), severe snowstorms have not abated: the number of severe regional snowstorms in the United States since 1960 has been more than twice that of the preceding six decades (Kunkel et al., 2013). When storms of this kind strike, workers exposed to harsh conditions often face absenteeism and reduced working hours, lowering the hours they actually work. These reductions may arise from constraints on labor supply, such as commuting difficulties, or from reductions in labor demand, such as business closures, or from both.

Building on classic studies that link climate change to economic activity (Mendelsohn et al., 1994; Deschênes and Greenstone, 2007), recent research has increasingly examined how weather events shape individual behavior. Most of this literature has focused on heat waves and their economic consequences, such as losses in gross domestic product (García-León et al., 2021; Wan and Wang, 2024), lower labor productivity (Somanathan et al., 2021; Han et al., 2024; ILO, 2019), and reduced working hours (Graff Zivin and Neidell, 2014). Particularly, Graff Zivin and Neidell (2014) found that workers adopt an adaptation strategy to extremely hot weather through temporal reallocation—reducing work hours and shifting time toward indoor leisure.

Despite mounting evidence that high temperatures impose significant economic losses,

relatively little attention has been paid to the impact of winter weather on the economy, including its effects on the labor market. Bloesch and Gourio (2015) examined short-run effects of low temperatures and snow on employment and retail sales, finding modest aggregate impacts but sizable damages in specific sectors such as utilities, construction, and hospitality. Liu and Hirsch (2021) showed that snowfall reduces U.S. working hours, with heterogeneous impacts across workers, occupations, industries, and regions. Their results suggest that workers adapt to winter weather in a manner similar to that during heat waves, primarily through reduced working hours.

Yet empirical evidence on alternative adaptation channels beyond temporal reallocation remains scarce. When workers must cut hours due to weather conditions, they are highly unlikely to fully offset lost income or utility costs, creating incentives to find alternative forms of adaptation that preserve working time. Spatial reallocation, the relocation of one's place of work from one location to another, could be a viable alternative. Furthermore, the spatial constraints of winter weather, driven by snowfall that reduces mobility and worsens outdoor work conditions, give workers greater motivation to adopt strategies that relax location-based barriers to work, highlighting the need to study adaptation through spatial reallocation.

In this paper, I show that telework serves as a spatial margin of adaptation to winter weather, available primarily to workers in telework-feasible occupations. The closest precedent is Li (2023), who uses the American Time Use Survey to show that workers with the flexibility to work from home shift work away from the workplace during extreme weather events. This paper differs in three respects. First, I treat snowfall as a continuous measure and estimate a dose-response relationship, recovering the intensity of the telework response rather than the effect of a binary shock. Second, I isolate winter weather rather than pooling many distinct extreme events into a single indicator, focusing specifically on snowfall as the relevant margin of disruption. Third, the longitudinal structure of the CPS permits individual fixed effects, providing sharper identification than

the pooled cross-sectional time-use data in which workers are observed only once.

This paper proposes remote work as an adaptation strategy that mitigates the impact of adverse weather by reallocating work spatially, from on-site workplaces to homes or other preferred locations. In recent years, remote work has become a prominent alternative work arrangement, particularly following the COVID-19 pandemic. Before 2020, only 6.5 percent of U.S. private-sector employees primarily worked from home (Pabilonia and Redmond, 2024). The pandemic, however, served as a large-scale, unplanned experiment in telework, as public health concerns and lockdowns prompted millions of workers and firms to adopt remote work arrangements. Even after the pandemic, the prevalence of telework remained well above pre-pandemic levels in many sectors.

In this research, I estimate the impact of winter weather on telework hours as evidence of telework's role as an adaptive response to weather conditions. In parallel, I examine its effect on on-site hours to assess the extent of substitution between the two work modes. The analysis combines individual-level data on telework hours from the 2022–2025 Current Population Survey (CPS) with daily weather records from the National Oceanic and Atmospheric Administration (NOAA). Using a two-way fixed effects (TWFE) estimation, I control for individual and temporal heterogeneity, and further estimate results by occupation-specific telework intensity as a proxy of telework feasibility of the occupation.

Beyond the baseline TWFE specification, I implement the continuous-treatment difference-in-differences framework of Callaway et al. (2024). This approach generalizes event-study designs to continuously varying treatments, allowing me to trace dynamic dose–response relationships between snowfall and telework hours and to reveal heterogeneous adaptation patterns beyond the average effects identified by the TWFE model.

This paper finds that greater snowfall increases telework hours, with this effect concentrated among occupations with higher telework feasibility: snowfall has little measurable effect in the lower-telework quartiles but becomes positive and sizable in the up-

per quartiles. Lower temperatures and higher wind speeds also raise telework hours, though the temperature estimates are weaker and less stable across specifications. Analyses using total and on-site hours confirm that additional snowfall reduces on-site work more strongly than total working hours, implying partial compensation of the loss in total working hours through telework. These findings suggest that telework partially offsets the reduction in on-site work caused by winter weather, though the degree of offset depends on occupational telework feasibility caused by winter weather.

These effects are economically meaningful, though moderate in magnitude. This reflects two features of the setting: most snowfall realizations are light enough that they are unlikely to disrupt commuting or alter the location of work, so the estimated response largely reflects exposure with little behavioral pressure to adapt, and the telework response is concentrated among the minority of workers in telework-feasible occupations. I therefore focus much of the analysis on the intensity of the response among exposed workers, rather than on full-sample average effects alone.

Furthermore, the estimated dose-response curve suggests a similar conclusion to that of the TWFE estimation. Individuals who experienced a larger amount of snow during their survey period tend to have more hours of telework. The result not only aligns with the estimated results from the TWFE regression but also demonstrates that the positive relationship between snow and telework is robust, after integrating the facts into the empirical model that the treatment begins at different times (staggered) with varying amounts for individual workers (continuous).

This paper makes several contributions to the growing literature on weather adaptation and labor economics. First, it introduces telework as an emerging behavioral adaptation to adverse weather conditions, particularly winter weather, which constrains mobility and outdoor activity. Second, it highlights the heterogeneity in telework feasibility across occupations as a key mechanism mediating workers' sensitivity to weather shocks. The increase in telework hours in response to snowfall is concentrated among occupa-

tions with higher telework feasibility, indicating that occupational characteristics play a significant role in shaping adaptive capacity. Third, the paper combines evidence from both a traditional two-way fixed effects model and a dynamic dose-response function with continuous treatment to provide robust evidence on the causal relationship between winter weather and telework behavior. Finally, the findings have broader implications for climate adaptation policy. Given that extreme weather events, such as floods and other natural disasters, are likely to impose spatial constraints, promoting telework infrastructure could serve as a general adaptation measure to maintain economic activity amid diverse environmental shocks.

The remainder of the paper is organized as follows. Section 2 presents the theoretical framework and derives its comparative statics. Section 3 describes the data sources used in the analysis and provides the summary statistics. Section 4 outlines the empirical strategy. Section 5 presents the main results on the impact of winter weather on telework hours. Section 6 examines the robustness of the main results, and Section 7 concludes the paper.

2 Theoretical Framework

2.1 Setup

In this section, I provide a simplified labor supply model to analytically investigate an individual's decisions on teleworking. The model describes the worker's decision problem on a single workday. Since teleworking is a combination of spatial (where to work) and temporal (how many hours to work) choices, the worker faces a two-step decision problem regarding the work arrangement each day:

- **Step 1 (Extensive Margin):** The worker chooses *where* to work (on site or from home).
- **Step 2 (Intensive Margin):** Conditional on the chosen location, the worker chooses

how many hours to work at that location.

Each worker solves the following utility maximization problem:

$$\max_{C,\,L} \; U(C, \tilde{L}) \tag{1}$$

subject to

$$C = w\left(H_s + \delta_k \cdot H_h\right), \tag{2}$$

$$L = T - H_s - H_h - \mathbf{1}(H_s > 0) \cdot T_c(W), \tag{3}$$

$$\tilde{L} = \varphi(W) \cdot L, \tag{4}$$

where C denotes consumption and L denotes leisure time. The worker allocates time between on-site working hours H_s, teleworking hours H_h, and leisure, subject to a total time endowment T. The utility function $U(C, \tilde{L})$ is strictly increasing and strictly concave in both arguments: $U_C > 0$, $U_{\tilde{L}} > 0$, $U_{CC} < 0$, and $U_{\tilde{L}\tilde{L}} < 0$.

Equation (2) is the budget constraint, where the worker earns a wage w per hour of on-site work. When teleworking, the worker's productivity is scaled by $\delta_k \in (0, 1)$, the relative telework efficiency for occupation k. This parameter represents how efficient occupation k is when working from home compared to on-site work, so that the effective wage from teleworking is $w \cdot \delta_k$.

Equation (3) is the time constraint. The term $T_c(W)$ captures the temporal cost of commuting, which is increasing in winter weather severity W, i.e., $T_c'(W) > 0$. The indicator function $\mathbf{1}(H_s > 0)$ ensures that commuting costs are incurred only when the worker travels to the workplace.

Equation (4) introduces $\tilde{L}$, the *effective leisure*, which captures the idea that weather conditions directly affect the quality of leisure time. The function $\varphi(W)$ is strictly positive and strictly decreasing in weather severity, with $\varphi(W) > 0$ and $\varphi'(W) < 0$: as winter

weather becomes more severe, the enjoyment derived from a given amount of leisure time decreases. Note that I do not consider the direct impact of weather conditions on consumption.

Finally, the wage w, the winter weather severity W, and the telework efficiency δ_k are all treated as exogenous[1]. For analytical tractability, the model excludes "hybrid working" within a day: on any given day, the worker either commutes to the workplace or works from home, but not both. Weekly observations that combine positive telework and on site hours are consistent with this assumption, since weekly hours aggregate daily location choices across the days of the reference week. Therefore, on each day the worker chooses one of the following:

1. **Commute and work at the workplace:** $H_s > 0$ and $H_h = 0$
2. **Work from home(telework):** $H_s = 0$ and $H_h > 0$

Case 1: Work at the workplace.

$$\max_{H_s} \; U(w \cdot H_s, \; \varphi(W) \cdot [T - H_s - T_c(W)]) \tag{5}$$

The first order condition is:

$$w \cdot U_C = \varphi(W) \cdot U_{\tilde{L}} \tag{6}$$

which implicitly defines the optimal on site working hours $H_s^*(w, W)$. The indirect utility function of this case is:

$$V_{\text{site}}(w, W) = U(w \cdot H_s^*, \; \varphi(W) \cdot [T - H_s^* - T_c(W)]) \tag{7}$$

[1]The model also abstracts from labor demand. In practice, severe winter weather may reduce the demand for labor through temporary business closures, canceled shifts, or reduced customer traffic, so that observed hours reflect both workers' choices and employers' responses. The framework here isolates the supply side decision in order to derive predictions about the choice of work location.

Case 2: Work from home.

$$\max_{H_h} \; U(w \cdot \delta_k \cdot H_h, \; \varphi(W) \cdot [T - H_h]) \tag{8}$$

The first order condition is:

$$w \cdot \delta_k \cdot U_C = \varphi(W) \cdot U_{\tilde{L}} \tag{9}$$

which implicitly defines $H_h^*(w, \delta_k, W)$. The indirect utility function is:

$$V_{\text{home}}(w, \delta_k, W) = U(w \cdot \delta_k \cdot H_h^*, \; \varphi(W) \cdot [T - H_h^*]) \tag{10}$$

2.2 Work Location Decision

Consequently, the worker decides to:

$$\text{Work at workplace if } V_{\text{site}} > V_{\text{home}}, \qquad \text{Work from home if } V_{\text{site}} < V_{\text{home}}.$$

Thus, the probability of working from home can be defined as:

$$P_{\text{WFH}} = P(V_{\text{home}} > V_{\text{site}}) \tag{11}$$

Without loss of generality, we can assume that P_{WFH} increases when V_{home} increases and/or V_{site} decreases.

2.3 Comparative Statics

Effect of weather on the probability of teleworking. We apply the envelope theorem to each indirect utility function. Since H_s^* satisfies the first order condition (6), the effect of W on V_{site} is obtained by differentiating only through the explicit dependence on W,

holding H_s^* fixed:

$$\frac{dV_{\text{site}}}{dW} = U_{\tilde{L}} \cdot \frac{\partial \tilde{L}}{\partial W}\bigg|_{H_s = H_s^*} = U_{\tilde{L}} \cdot \Big[\varphi'(W) \cdot L_s^* \; - \; \varphi(W) \cdot T_c'(W) \Big] \tag{12}$$

where $L_s^* = T - H_s^* - T_c(W) > 0$ is the optimal leisure time under on site work. Since $U_{\tilde{L}} > 0$, and both terms inside the brackets are negative—$\varphi'(W) < 0$ with $L_s^* > 0$, and $\varphi(W) > 0$ with $T_c'(W) > 0$—we have:

$$\frac{dV_{\text{site}}}{dW} < 0$$

This result reflects two distinct channels through which worsening weather reduces the value of on-site work:

1. **Commuting cost effect** $(-\varphi(W) \cdot T_c'(W))$: Severe weather increases commuting time, directly reducing the leisure time available to the on-site worker.

2. **Leisure depreciation effect** $(\varphi'(W) \cdot L_s^*)$: Severe weather reduces the quality of remaining leisure time.

Similarly, applying the envelope theorem to V_{home} using the first order condition (9):

$$\frac{dV_{\text{home}}}{dW} = U_{\tilde{L}} \cdot \varphi'(W) \cdot L_h^* < 0 \tag{13}$$

where $L_h^* = T - H_h^* > 0$. The home worker is also affected by weather through the leisure depreciation channel, but crucially, the commuting cost channel is absent.

Since both V_{site} and V_{home} decrease with worsening weather, the net effect on P_{WFH} depends on which decreases more. The key observation is that V_{site} is subject to *both* the commuting cost effect and the leisure depreciation effect, whereas V_{home} is subject *only* to the leisure depreciation effect. The additional negative term $-\varphi(W) \cdot T_c'(W)$ in equation (12) ensures that V_{site} declines by more than V_{home}, all else equal. Therefore:

$$\frac{dP_{\text{WFH}}}{dW} > 0 \tag{14}$$

Equation (14) establishes the extensive margin response to winter weather severity: as weather conditions worsen, workers are more likely to choose telework over on-site work, thereby increasing the probability of working from home.

Effect of telework efficiency on the probability of teleworking. Let $k = 1, 2, 3, \ldots, K$, and assume $0 < \delta_1 < \delta_2 < \cdots < \delta_K < 1$. A larger δ_k implies a greater telework efficiency. Workers in occupation m return more output when teleworking compared to workers in occupation $m - n$ $(0 < n \leq m - 1)$. The worker with occupation k earns an "effective wage" of $w \cdot \delta_k$ when working from home.

Applying the envelope theorem to V_{home} with respect to δ_k, using the first order condition (9):

$$\frac{dV_{\text{home}}}{d\delta_k} = \underset{(+)}{U_C} \cdot \underset{(+)}{w} \cdot H_h^* > 0 \tag{15}$$

Since V_{site} does not depend on δ_k, an increase in telework efficiency raises V_{home} while leaving V_{site} unchanged. Consequently:

$$\frac{dP_{\text{WFH}}}{d\delta_k} > 0 \tag{16}$$

That is, workers in occupations with higher telework efficiency are more likely to work from home.

Effect of weather on on-site working hours. For workers who choose to work on site, we can express on-site working hours from the time constraint as:

$$H_s = T - T_c(W) - L \tag{17}$$

Totally differentiating with respect to W:

$$\frac{dH_s^*}{dW} = \underset{(+)}{-T_c'(W)} - \frac{dL^*}{dW} \tag{18}$$

The first term, $-T_c'(W) < 0$, is the commuting cost effect: worsening weather increases commuting time, which mechanically reduces the time available for on-site work, pushing H_s downward.

For the second term, note that as W increases, $\varphi(W)$ decreases, which lowers the marginal utility of leisure ($\varphi(W) \cdot U_{\tilde{L}}$). The worker optimally substitutes away from leisure toward work, so $dL^*/dW < 0$. This refers to the leisure depreciation effect: since bad weather makes leisure less enjoyable, the worker is motivated to work more, pushing H_s upward via $-dL^*/dW > 0$.

The two effects work in opposite directions. The commuting cost effect discourages on-site work, while the leisure depreciation effect encourages it. Without further parametric restrictions, the net effect of winter weather on on-site working hours is ambiguous:

$$\frac{dH_s^*}{dW} = \underbrace{-T_c'(W)}_{\substack{\text{commuting cost} \\ \text{effect } (<0)}} \quad \underbrace{-\frac{dL^*}{dW}}_{\substack{\text{leisure depreciation} \\ \text{effect } (>0)}} \quad \gtrless \quad 0 \tag{19}$$

The sign depends on the relative magnitudes of the two effects, which is ultimately an empirical question[2].

[2]Labor demand responses reinforce this ambiguity. For most occupations, severe winter weather plausibly reduces labor demand, for example, through workplace closures or reduced customer traffic, placing additional downward pressure on on-site hours; a small set of winter-specific occupations, such as snow removal, is a notable exception. Because realized hours reflect both supply and demand responses, the sign of the net effect cannot be determined by the supply side model alone.

3 Data

3.1 Weather Data

The weather data employed in this study are provided by the National Oceanic and Atmospheric Administration(NOAA). The data archived by the National Centers for Environmental Information (NCEI) contains comprehensive and integrated historical weather datasets. One of the datasets is the Global Historical Climatology Network Daily (GHCN-Daily) dataset, which combines daily weather measurements from approximately 30 distinct sources, containing variables such as precipitation, daily high and low temperatures, wind speed, snowfall, and snow depth (Menne et al., 2012a,b). These data are collected from over 90,000 terrestrial stations worldwide, with all temperature readings measured in Celsius, which I convert to Fahrenheit for the analysis. In this paper, I use the daily average temperature, wind speed, and snowfall to represent winter weather. The average temperature is the mean of the daily minimum and maximum temperatures. Since the CPS records telework hours for the reference week, all weather variables used in the analyses, namely average temperature, wind speed, and snowfall, are aggregated to that same week by averaging the daily observations over the seven days of the reference week.

Constructing an effective identification strategy for the impact of snow on remote work requires selecting a snow-related variable that best captures the relevant margin of disruption. Snowfall measures the amount of newly fallen snow over the observation interval, whereas snow depth measures the total amount of snow remaining on the ground at the time of observation. I aggregate daily snowfall (snow depth) to the CPS reference week exposure window by averaging daily snowfall (snow depth) over the seven-day period.

Although both snow measures capture snow-related winter conditions, they represent conceptually distinct aspects of snow exposure. Snow depth reflects persistent snow accumulation on the ground and can capture lingering mobility frictions associated with

prolonged winter conditions, such as reduced accessibility around residences, sidewalks, and parking areas. However, snow depth is measured on undisturbed ground surfaces rather than on treated road surfaces and therefore does not directly represent the amount of snow remaining on roads after plowing or salting. This distinction is particularly relevant in areas with greater winter weather preparedness, where substantial snow accumulation on untreated ground can coexist with relatively rapid restoration of commuting routes.

Snowfall, by contrast, captures newly occurring snow shocks during the reference week. This measure is more closely aligned with the short-run disruptions that prompt immediate shifts from on-site to remote work: transit delays and cancellations, school and childcare closures, temporary workplace schedule adjustments, and the acute deterioration of road conditions before plowing can restore normal commuting routes. Because this paper examines whether adverse winter weather leads to short-run changes in the location of work, I use average daily snowfall during the reference week as the primary snow-related weather measure. I use snow depth as an alternative measure in robustness analyses to assess whether the results are robust to characterizing snow exposure as persistent accumulation rather than newly occurring snowfall. Summary statistics of weather variables can be found in the Appendix.

3.2 Telework Hours Data

The telework hours data used in this study are drawn from the IPUMS Current Population Survey (Flood et al., 2024), which harmonizes microdata from the monthly U.S. labor force survey. Respondents of the Current Population Survey (CPS) are asked to report information about events that occurred during the previous week (the "reference week"), which is usually the seven-day calendar week that includes the 12th of the month. In November and December, however, the reference week is sometimes moved one week

earlier to avoid survey operations during major holiday periods[3]. Accordingly, when merging the CPS with the weather data, I match the daily weather observations to the exact reference week dates published by the BLS for each month, rather than relying on the week of the 12th rule.

The CPS provides rich demographic information and is helpful for this study because it records how much the individual spent on remote work during the reference week. I focus on individuals residing in Metropolitan Statistical Areas (MSAs) identified in the CPS, restricting the sample to individuals with valid occupation codes. The baseline estimation sample covers 193 MSAs[4]. Because the telework hours data is available from October 2022, the dataset's sample period is October 2022 to March 2025. Moreover, this paper limits the sample period from January to March and October to December each year to concentrate on the impact of winter weather.

A merit of using the CPS data lies in its longitudinal design, which enables tracking individuals within the same season across two successive years. This is made possible by the survey's unique structure: respondents are interviewed for four months in a row, take an eight-month break, and are then surveyed again for another four-month period. Consequently, the dataset provides repeated observations for individuals across two consecutive winters. Because the CPS is an address-based survey, households that move are not followed to their new residence. Individuals, therefore, appear repeatedly in the panel only if they remain at the same address, and hence in the same MSA, so the within-person variation that identifies the individual fixed effects comes entirely from residentially sta-

[3]The exact reference week dates for each month are published by the U.S. Bureau of Labor Statistics at https://www.bls.gov/cps/definitions.htm. Within the sample period, the reference weeks for December 2022, November and December 2023, and November and December 2024 were moved one week earlier and do not contain the 12th.

[4]The IPUMS CPS extract identifies 309 MSAs, of which 252 are matched to GHCN Daily weather stations. Estimation further drops observations with missing weather variables during the reference week, as well as individuals observed only once, whose observations are absorbed by the individual fixed effects. Of the 59 MSAs that exit the sample at this stage, 58 exit because all of their observations lack complete weather data for the relevant reference weeks, and one exits through the absorption of single observation individuals. The 193 MSAs in the baseline sample therefore reflect the availability of complete weather station coverage rather than any additional sample restriction.

ble workers. This feature is well-suited to the question at hand: the analysis concerns short-run adjustments in the location of work, not migration responses, and the fixed residential location ensures that variation in a worker's weather exposure arises solely from the timing of weather events rather than from moves across areas [5].

Throughout the analysis, I classify occupations by their telework feasibility, which serves as the central dimension of heterogeneity. I measure telework feasibility using the average weekly telework hours within each occupation, computed over all workers in that occupation across the full October 2022 to March 2025 sample. Occupations are defined using the detailed occupation classification in the CPS (the IPUMS-CPS OCC variable), which distinguishes occupations at the four-digit level[6]. I rank these occupations by their average telework hours and group them into quartiles, with the first quartile containing occupations with the lowest average telework hours and the fourth containing those with the highest. I additionally isolate the top decile of occupations by average telework hours to capture the most telework-intensive jobs. This occupation-level measure is intended to proxy for the technical and organizational scope for remote work, a structural feature of the occupation rather than a worker's realized telework in any particular week.

Table 1 presents summary statistics for the full sample and for selected occupational subgroups. Several patterns merit attention. First, the average telework hours vary substantially across occupational groups: workers in the 1st quartile report only 0.23 hours of telework per week, while those in the top decile report 19.32 hours, underscoring the wide disparity in telework feasibility across occupations.

Second, while total working hours are slightly higher for workers in the 4th quartile and top decile (38.28 and 37.85, respectively) compared to those in the 1st quartile (32.69), the composition of these hours differs markedly. Workers in the 1st quartile perform

[5]The cost is that workers who move during the survey period, for any reason, contribute at most one observation; given the short panel window and the transitory nature of weather shocks, weather-induced moves within this horizon are unlikely to be a significant source of attrition.

[6]The sample contains 526 distinct four-digit occupations. Because the sample period falls entirely within the 2020 Census occupational classification, these codes are consistent throughout the sample.

Table 1. Summary Statistics

	All (N = 467,103)		1st Quartile (N = 109,475)		4th Quartile (N = 128,127)		Top Decile (N = 45,777)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Working hours								
Actual working hours	35.46	14.67	32.69	15.59	38.28	13.08	37.85	12.28
Telework hours	5.92	13.17	0.23	2.66	15.07	17.70	19.32	18.25
On-site working hours	29.53	17.73	32.46	15.69	23.21	19.30	18.52	18.58
Telework ratio	0.16	0.34	0.01	0.08	0.40	0.43	0.52	0.44
Weather conditions								
Average temperature (°F)	51.74	13.93	52.20	13.93	51.39	13.79	51.19	13.66
Snowfall (inches)	0.07	0.25	0.07	0.25	0.07	0.25	0.07	0.26
Snow depth (inches)	0.29	1.23	0.27	1.19	0.29	1.22	0.28	1.19
Average wind speed (miles per hour)	7.76	2.60	7.72	2.60	7.77	2.59	7.77	2.58
Regions and divisions								
New England	0.05	0.23	0.05	0.21	0.06	0.24	0.06	0.24
Middle Atlantic	0.05	0.22	0.05	0.22	0.04	0.21	0.04	0.20
East North Central	0.16	0.37	0.16	0.37	0.16	0.37	0.16	0.37
West North Central	0.05	0.22	0.05	0.21	0.05	0.22	0.05	0.22
South Atlantic	0.19	0.39	0.19	0.39	0.19	0.39	0.19	0.39
East South Central	0.05	0.21	0.05	0.22	0.04	0.20	0.04	0.19
West South Central	0.16	0.36	0.17	0.37	0.15	0.35	0.14	0.35
Mountain	0.08	0.28	0.08	0.27	0.08	0.28	0.09	0.28
Pacific	0.20	0.40	0.20	0.40	0.22	0.41	0.23	0.42
Demographics								
Age	42.09	14.44	39.26	15.34	44.04	13.15	42.08	12.62
Female	0.47	0.50	0.32	0.47	0.44	0.50	0.42	0.49
No HS diploma	0.07	0.26	0.19	0.40	0.01	0.09	0.00	0.06
HS graduate	0.39	0.49	0.60	0.49	0.20	0.40	0.17	0.38
Bachelor's degree	0.37	0.48	0.19	0.39	0.52	0.50	0.58	0.49
Graduate degree	0.16	0.37	0.02	0.13	0.26	0.44	0.24	0.43

Notes: Working hours are at the weekly level. On-site working hours equal the actual working hours minus telework hours. The telework ratio is measured as the share of telework hours in actual working hours for each worker; the number of observations for this variable is smaller than the full sample because the ratio is undefined when actual working hours equal zero. Weather conditions are the weekly average of daily observations; the number of observations for weather variables may differ from the reported sample size due to missing weather station data. The 1st quartile group includes workers in occupations with the lowest quartile of annual average telework hours across all occupations, while the 4th quartile group comprises workers in occupations with the highest quartile. The top decile group comprises workers in occupations with the highest annual average telework hours. Statistics are weighted using the CPS final person weight (wtfinl).

nearly all of their work on site (32.46 out of 32.69 hours), whereas those in the top decile split their hours roughly equally between telework (19.32) and on-site work (18.52). This compositional difference is central to the analysis, as it implies that the scope for substitution between on-site and remote work in response to weather shocks is substantially greater for higher-quartile occupations.

Third, weather conditions are largely similar across occupational groups, with average temperatures around 50°F and snowfall averaging 0.07 inches. This similarity is expected, as weather is determined by geographic location rather than occupation. The near-uniformity of weather exposure across groups helps alleviate concerns that the heterogeneous effects documented in the regression analysis are driven by differential expo-

sure to winter weather rather than differential responsiveness.

Finally, the demographic composition of occupational groups reveals notable differences. Workers in the 1st quartile are younger on average (39.26 years) and more likely to be male (68 percent), and educational attainment in this group is lower: 19 percent have no high school diploma and 2 percent hold a graduate degree. In contrast, educational attainment in the top decile is considerably higher, with 58 percent holding a bachelor's degree and 24 percent holding a graduate degree. Educational attainment in the sample is higher than in the U.S. adult population as a whole. This primarily reflects the restriction of the sample to employed individuals, whose educational attainment exceeds that of the general population, and to a lesser extent, the restriction to metropolitan area residents. These compositional differences further motivate the use of individual fixed effects in the baseline specification, which absorb time-invariant individual characteristics that may be correlated with both occupational choice and telework propensity. Moreover, the heterogeneity observed in these demographics raises the question of whether the telework pattern could also differ across other demographic compositions.

4 Empirical Strategy

4.1 Baseline Model: Two-Way Fixed Effects Regression

To examine the effect of cold weather on remote work, this study adopts a two-way fixed effects (TWFE) model. Panel fixed effects designs that exploit plausibly exogenous variation in weather are standard in the climate-economy literature (Deschênes and Greenstone, 2007; Deschênes and Greenstone, 2011), as they control for both unobserved unit heterogeneity and common temporal shocks. The proposed econometric specification is:

$$H_{ijt} = \beta_0 + \beta_1 T_{jt} + \beta_2 W_{jt} + \beta_3 S_{jt} + \alpha_i + \gamma_t + \epsilon_{ijt} \tag{20}$$

Where H_{it} denotes the total telework hours of individual i in a reference week of each month, living in the metropolitan statistical area (MSA) j at year-month t. The variable T_{jt} represents the weekly average of the daily average temperature in MSA j at time t, while W_{jt} and S_{jt} represent the weekly average wind speed and snowfall, respectively. Therefore, the estimated coefficients of β_1, β_2, and β_3 are expected to capture the impact of winter weather on telework hours[7].

The model includes the parameter γ_t to account for the time-fixed effect, specifically a year-month fixed effect for the analysis. This parameter is expected to capture the impact of macroeconomic conditions and weather seasonality on telework hours. The parameter α_i denotes the individual fixed effect, which absorbs time-invariant individual characteristics such as skill, baseline preferences for remote versus on-site work, and tolerance for cold weather. The error term is denoted by ϵ_{it}, reflecting unobserved factors that influence individuals' realized working hours. Because individual fixed effects control for time-invariant individual characteristics, and time-fixed effects capture shocks common to all individuals at a given time, the remaining error term is assumed to reflect short-term unobserved variations across individuals, such as day-to-day fluctuations in physical condition, and is expected to be uncorrelated with the independent variables.

To allow for nonlinear temperature effects, I also estimate an alternative specification that replaces the continuous temperature measure with a set of temperature-bin indicators. This approach follows a common strategy in the climate-economy literature, where the relationship between temperature and economic outcomes is often modeled flexibly rather than imposing a constant marginal effect across the full temperature distribution

[7]Because the CPS records hours actually worked, the estimated coefficients reflect the combined influence of labor supply and labor demand. For instance, snowfall may discourage commuting (a supply-side response) while also prompting temporary business closures or canceled shifts (a demand-side response), and the two cannot be separately identified with data on realized hours alone. The theoretical framework in Section 2 isolates the supply channel, and the concentration of telework responses among telework-feasible occupations is consistent with a supply-side mechanism; the estimates should nonetheless be interpreted as overall effects on realized work rather than as pure supply responses. Relatedly, the observed shift toward telework may reflect decisions by workers, by employers, or both; I use the term adaptation to describe the reallocation of work across locations without attributing it to either side alone.

(Schlenker and Roberts, 2009; Deschênes and Greenstone, 2011; Burke et al., 2015).

Specifically, I estimate the following specification:

$$Y_{ijt} = \beta_0 + \beta_1 S_{jt} + \beta_2 W_{jt} + \sum_{b \neq b_0} \beta_3^b TB_{jt}^b + \alpha_i + \gamma_t + \varepsilon_{ijt}. \tag{21}$$

Here, the main difference from Equation (20) is that the effect of temperature is captured by multiple temperature bins. Each temperature bin TB_{jt}^b counts the number of days in the reference week of year-month t that fall into the temperature criteria of the bin b. For instance, β_3^1 captures the effect of temperature for an additional day that falls into the criteria of temperature bin 1, or an increase in TB^1 by one. To avoid multicollinearity, a specific bin (b_0) is intentionally omitted from the estimation.

Relative to the continuous temperature specification, the bin specification does not require the marginal effect of temperature to be constant across the winter temperature distribution. This flexibility is useful because the behavioral response to cold weather may be concentrated at very low temperatures: moderate declines in temperature may have limited effects on work arrangements, whereas extremely cold conditions may increase commuting costs, raise the likelihood of school or office disruptions, or otherwise make telework more attractive.

4.2 Dose-Response Extension of a Staggered DiD Event-Study Framework

To complement the baseline two-way fixed effects (TWFE) specification, I further employ the continuous-treatment difference-in-differences framework developed by Callaway et al. (2024). This framework extends conventional event-study designs to settings in which both treatment timing and treatment intensity vary continuously across observational units. In the context of this study, the intensity of winter weather — measured by snowfall — is treated as a continuous treatment rather than a binary exposure. This approach

enables the estimation of *dose-response functions* that track how telework hours evolve over time for the individuals experiencing varying levels of snowfall.

Formally, the framework identifies group-time-dose-specific average treatment effects,

$$ATT(g, t, d) = E[Y_t(d) - Y_t(0) \mid G = g, D = d], \tag{22}$$

where g denotes the first period in which an individual experiences a given level of snow, t denotes the observation period, and d represents the intensity of the treatment. The effect is defined relative to the untreated potential outcome $Y_t(0)$, which I approximate using a comparison group composed of workers not yet exposed to snow within the same winter, together with workers who experience no snow during that winter. By aggregating these effects over both event time and dose intervals, I obtain an event-study-type dose-response function,

$$ATT^{es}_{e_1,e_2}(d) = \frac{1}{e_2 - e_1 + 1} \sum_{e=e_1}^{e_2} E[ATT(G, G + e, d) \mid G + e_2 \in [2, T], D = d], \tag{23}$$

which summarizes the average response to a given level of treatment over a specific post-exposure window.

The function $ATT^{es}_{e_1,e_2}(d)$ can be interpreted as a *dynamic dose-response curve*: for each level of treatment intensity d, it shows how the average telework response evolves across the two different posttreatment event times e_1 and e_2, such that $0 \leq e_1 \leq e_2$. Callaway et al. (2024) exploit the two different event windows to compare the dose responses between the short-run and the long-run. However, to focus on the immediate response of the individuals to the snow and reduce the possibility of utilizing other adaptation strategies in the long run, I estimate the function with only one event window where $e_1 = 0$ and $e_2 = 1$. Therefore, in this research, the function $ATT^{es}_{e_1,e_2}(d)$ stands for the average treatment effect of being exposed to the level of snowfall d among all units that have been treated for at least 1 month, averaged over event times 0 (the month in which

the individual first experiences snow that winter) and 1 (the following month). A positively sloped dose-response curve indicates that telework increases with the intensity of snowfall. Conversely, a flatter or negative slope would suggest limited adaptability or constraints on telework substitution under snowy weather.

Identifying assumptions. The continuous treatment framework identifies the dose response function under a no anticipation condition and two versions of a parallel trends assumption that differ in what they license me to interpret causally.

No anticipation requires that telework hours do not respond to snowfall before it occurs. Because the realized intensity of a given winter weather event is difficult to forecast more than a few days in advance, and because the CPS reference week is fixed by survey design rather than chosen by the worker, anticipatory adjustment of telework to a specific future snowfall is implausible in this setting.

A *weak parallel trends* assumption identifies, at each realized snowfall intensity d, the average treatment effect for the workers who actually experienced that intensity, $ATT(d \mid d) = E[Y_t(d) - Y_t(0) \mid D = d]$. It requires that, absent snow exposure, the average telework hours of workers who experienced snowfall of intensity d would have evolved in parallel with those of the comparison workers who had not yet been exposed to snow within the same winter, or who experienced no snow that winter. Under this assumption, $ATT(d \mid d)$ recovers the causal effect of being exposed to dose d relative to no exposure, for the group that received that dose. This is a level object: it speaks to the effect of snow at a given intensity, but it does not by itself describe how a given worker's telework would respond were her snowfall marginally increased. The aggregated counterpart estimated here is $ATT^{es}_{e1,e2}(d)$ in equation (23).

Interpreting the slope of the estimated curve as a causal dose-response requires the stronger condition called strong parallel trends. Plotting $ATT(d \mid d)$ against d and reading its slope conflates the causal response to a change in dose with selection into dose:

each point on the curve pertains to a different group of workers, and those who happen to receive heavier snowfall may differ systematically from those who receive lighter snowfall. Strong parallel trends rule out such selection by requiring that the path of potential telework hours under any fixed dose d would have been the same across all groups, regardless of the snowfall intensity each group actually realized. This assumption is practically untestable. The following two considerations make it more plausible here than in a typical setting where treatment intensity is endogenously chosen by agents, though neither is decisive. First, snowfall intensity is not a behavioral choice: conditional on metropolitan area and year-month, the amount of snow a worker experiences during the reference week is governed by meteorological factors that are plausibly orthogonal to that worker's underlying returns to telework, which limits selection into dose on the basis of telework adaptability. Second, the level of the dose-response function is identified under the weaker assumption alone, so the finding that telework responds positively to snow exposure does not hinge on strong parallel trends. I therefore treat the estimated "levels" as my primary causal object, while carefully reading the upward "slope" of the curve as suggestive evidence of how the response scales with intensity rather than as a sharply defined causal gradient.

Limitations. Several features of this design warrant emphasis. First, snowfall is a recurrent rather than an absorbing treatment. Because I summarize each worker's exposure by the snowfall intensity at first exposure, a worker recorded at a low first exposure dose may face heavier snow in a later week, introducing measurement error in dose that the framework does not address; restricting attention to the immediate post-exposure window mitigates but does not eliminate this concern. Second, the comparison group combines workers who are not yet exposed to snow during a given winter with those who experience no snow that winter. The latter are likely to be disproportionately drawn from warmer regions, so part of the cross-group variation reflects climate rather than

the timing of a common shock; the metropolitan area and year-month structure absorbs fixed locational differences, though it would not absorb region-specific seasonal patterns in telework if such patterns happened to align with snow exposure. Relatedly, because the CPS does not follow movers, all within-person comparisons are made among workers who remain in the same MSA throughout their survey participation (see Section 3.2). Third, the level and the slope of the dose-response function rest on different identifying assumptions. The level of the function at a given dose, that is, the effect of experiencing snowfall of intensity d relative to no exposure, is identified under the weaker parallel trends assumption alone. Reading the slope of the curve as the causal effect of additional snowfall, by contrast, requires the strong parallel trends assumption discussed above, because each point on the curve pertains to a different group of workers. I therefore base the paper's central claims on the two-way fixed effects estimates and on the level of the dose-response function, and interpret the slope as suggestive evidence on how the response scales with intensity.

5 Results

This section presents the main empirical results. I first establish that winter weather, and snowfall in particular, increases telework hours, with the response concentrated in occupations where remote work is feasible. I then show that this increase offsets much of the weather-induced decline in on-site hours, examine how the response scales with snowfall intensity, and assess whether alternative classifications organize the heterogeneity as systematically as occupational telework feasibility. The section closes by decomposing the response into extensive and intensive margins.

5.1 Baseline Effects on Telework Hours

Table 2 presents regression results estimating the effect of winter weather conditions – cold temperature, wind, and snow – on workers' telework hours. Panel A uses weekly average temperature as a continuous independent variable, while Panel B adopts temperature bins. Specifically, the low-temperature bin counts the number of days in each reference week with an average temperature below 20°F, the mid-temperature bin includes days between 20°F and 40°F, and the high-temperature bin covers days between 40°F and 60°F. Days above 60°F are omitted to avoid multicollinearity. Both panels additionally include the weekly average wind speed and snowfall. Estimation results for a finer bin specification are provided in the Appendix.

Column (1) reports results for the full sample. In Panel A, a one-degree decrease in temperature is associated with a 0.011-hour increase in telework, while an additional inch of snowfall increases telework hours by 0.246. Wind speed is also positively associated with telework hours, with a one-mile-per-hour increase corresponding to a 0.024-hour increase. In Panel B, however, the evidence for temperature effects weakens once temperature is modeled using bins: the estimates provide limited evidence of a nonlinear temperature response and do not reveal a clear monotonic relationship between colder days and telework hours. By contrast, both wind speed and snowfall remain positive and statistically significant, with the snowfall coefficient remaining nearly unchanged. Taken together, the full-sample results are driven more consistently by snow-related and windy conditions than by a robust nonlinear temperature-bin pattern.

The full-sample snowfall coefficient, while precisely estimated, is moderate in magnitude: an additional inch of snowfall raises telework by 0.246 hours, and moving from no snowfall to the 90th percentile among positive-snowfall weeks implies an increase of roughly 0.265 hours (Table 3), about 4 to 5 percent of mean weekly telework. Two features of the setting account for this magnitude. First, most snowfall realizations in the sample are light: among reference weeks with positive snowfall, the median aver-

Table 2. The impact of winter weather on telework hours across occupational groups

	Telework hours					
	Full Sample	1st Quartile	2nd Quartile	3rd Quartile	4th Quartile	Top 10 percent
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Weekly average temperature						
Temperature	-0.011***	0.000	-0.010***	-0.010*	-0.020**	-0.023
	(0.003)	(0.002)	(0.003)	(0.005)	(0.010)	(0.015)
Wind speed	0.024**	0.001	0.006	0.023	0.053*	0.106**
	(0.011)	(0.006)	(0.012)	(0.019)	(0.029)	(0.045)
Snowfall	0.246***	-0.031	0.029	0.496***	0.472**	0.989***
	(0.070)	(0.049)	(0.055)	(0.122)	(0.202)	(0.285)
Panel B: Temperature bin						
Low-temperature bin (20°F or below)	0.040	-0.012	0.066*	0.110*	0.024	0.115
	(0.029)	(0.019)	(0.037)	(0.058)	(0.074)	(0.132)
Mid-temperature bin (20°F to 40°F)	0.036*	-0.020*	0.031*	0.040	0.063	0.034
	(0.020)	(0.010)	(0.018)	(0.035)	(0.050)	(0.090)
High-temperature bin (40°F to 60°F)	0.010	-0.013*	0.013	0.025	-0.015	-0.056
	(0.013)	(0.008)	(0.012)	(0.029)	(0.031)	(0.054)
Wind speed	0.023**	0.001	0.006	0.022	0.048	0.096**
	(0.011)	(0.006)	(0.012)	(0.020)	(0.030)	(0.046)
Snowfall	0.250***	-0.031	0.032	0.497***	0.432**	0.876***
	(0.073)	(0.048)	(0.056)	(0.118)	(0.211)	(0.272)
Observations	467,103	109,475	90,939	116,350	128,127	45,777

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table reports coefficients from regressions of telework hours on winter weather measures. The full estimation sample includes workers in 193 metropolitan areas. The dependent variable is the number of hours worked from home during the CPS reference week. Panel A uses continuous measures of average temperature, wind speed, and snowfall, reported in degrees Fahrenheit, miles per hour, and inches, respectively. Panel B replaces the continuous temperature measure with temperature-bin indicators, with 60°F or above as the omitted category. Column (1) uses the full estimation sample, while Columns (2)–(5) restrict the sample to occupation-based quartiles defined by average telework hours. Column (6) restricts the sample to occupations in the top decile of average telework hours. All specifications include individual and year-month fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

age daily snowfall is only 0.14 inches (Table A1). Snowfall of this intensity is unlikely to impede commuting or to prompt a change in the location of work, so the behavioral response is plausibly weak over much of the observed snowfall distribution. Because the linear specification weights marginal effects toward the region of the distribution where snowfall variation is most abundant (Yitzhaki, 1996), the estimated per-inch coefficient primarily reflects responses to such light snowfall and understates the response to the heavier events that genuinely disrupt commuting. This interpretation is consistent with

the dose-response estimates presented below, which show larger telework responses at higher snowfall intensities. Second, the response is highly concentrated across occupations. The snowfall coefficient is statistically indistinguishable from zero in the first two telework-feasibility quartiles but rises to 0.989 in the top decile, roughly four times the full-sample estimate. Averaging across occupations with little scope for remote work and those with substantial scope, therefore, dilutes the full-sample coefficient. Read in this light, the moderate average effect reflects not a weak behavioral response among those able to telework, but the predominance of light snowfall events in the identifying variation and the concentration of telework capacity in a subset of occupations. This motivates the emphasis in the remainder of the analysis on the dose-response relationship and on heterogeneity across telework-feasible occupations, where the response among exposed workers can be isolated from the diluting influence of untreated weeks and unresponsive occupations.

Columns (2) through (5) report results by occupational quartiles, ranked by average annual telework hours within each occupation. Column (2) corresponds to occupations in the lowest quartile and Column (5) to those in the highest, so the classification captures differences in the scope for remote work across occupations, with higher quartile occupations more amenable to telework. The temperature estimates provide some evidence that colder conditions increase telework hours in more telework-intensive occupations, but the pattern is not uniform across specifications. In Panel A, the linear specification implies that colder conditions raise telework hours in the upper quartiles. The bin estimates in Panel B show no clear monotonic pattern in cold weather. By contrast, snowfall displays a more systematic occupational gradient: it has little measurable effect in the first two quartiles but becomes positive and sizable in the third and fourth. Wind speed shows a weaker version of the same pattern, with the largest responses concentrated among the most telework-intensive occupations. Overall, these results suggest that weather-induced increases in telework are concentrated in occupations where remote work is more feasible,

with the evidence clearest for snowfall.

The snowfall estimates are not perfectly monotonic across occupational quartiles: the third-quartile coefficients slightly exceed those of the fourth quartile in both panels. I therefore interpret the quartile results as evidence of a broad telework feasibility gradient rather than a strictly monotonic one. This interpretation is reinforced by Column (6), which further isolates the top decile of occupations by average telework hours. For this group, the estimated snowfall response is substantially larger than in the broader quartile groups, with coefficients of 0.989 in Panel A and 0.876 in Panel B, roughly four times the corresponding full sample estimate. Wind speed likewise shows its largest estimates in the top decile, whereas the temperature estimates remain less systematic. Together, these results suggest that workers in occupations with the greatest scope for remote work exhibit the strongest telework response to snow-related and windy conditions, while the evidence for temperature-driven responses is more limited.

Because weather variables are measured in different physical units, the raw coefficients in Table 2 are not directly comparable across weather dimensions. To facilitate comparison, Table 3 reports the implied effects of distribution-based winter weather shifts. This exercise is motivated by previous studies that define weather shocks relative to empirical weather distributions rather than relying only on raw physical units. For example, Shah and Steinberg (2017) define rainfall shocks using district-specific percentile cutoffs, while Kim et al. (2022) use standardized anomalies and percentile-based components to combine different dimensions of severe weather into a common index. In a similar spirit, I translate the baseline coefficient estimates into predicted changes in telework hours associated with empirically meaningful shifts in each weather variable.

Specifically, I consider a decline in weekly average temperature from the median to the 10th percentile, an increase in weekly average wind speed from the median to the 90th percentile, and an increase in snowfall from zero to the 90th percentile among positive-snowfall reference weeks. This approach does not redefine the treatment variables used

Table 3. Economic magnitudes of distribution-based winter weather shifts

Weather dimension	Distribution-based shift	Weather change	Implied change in telework hours	95% confidence interval
Temperature	Median → 10th percentile	−19.63°F	0.219	[0.094, 0.344]
Wind speed	Median → 90th percentile	3.61 mph	0.085	[0.006, 0.165]
Snowfall	No snow → 90th percentile among positive-snowfall weeks	1.07 inches	0.265	[0.115, 0.414]

Notes: This table reports post-estimation implied effects of distribution-based winter weather shifts on telework hours. The implied effects are calculated by multiplying each weather variable's baseline coefficient estimate by its corresponding change. For temperature, the shift is defined as a decline from the median to the 10th percentile of weekly average temperature. For wind speed, the shift is defined as the increase from the median to the 90th percentile of weekly average wind speed. For snowfall, because the distribution has a mass point at zero, the shift is defined as moving from no snowfall to the 90th percentile of average daily snowfall among positive-snowfall reference weeks. Confidence intervals are based on standard errors from the baseline regression, clustered at the MSA level.

in the baseline regressions; rather, it provides a post-estimation magnitude comparison that places the estimated effects on a common outcome scale. The results indicate that moving from the median to the 10th percentile of weekly average temperature is associated with a 0.219-hour increase in telework hours. Moving from the median to the 90th percentile of wind speed is associated with a smaller increase of 0.085 hours. For snowfall, moving from no snowfall to the 90th percentile among positive-snowfall weeks is associated with a 0.265-hour increase in telework hours. Thus, under this distribution-based comparison, snowfall and colder temperatures imply larger changes in telework hours than wind speed.

5.2 Substitution Between On-Site and Remote Work

Table 4 extends the analysis by examining whether winter weather affects the allocation of work hours between total work and on-site work. Panel A uses actual weekly working hours as the dependent variable, while Panel B uses on-site working hours, defined as total working hours net of telework hours. In the full sample, actual working hours decline with snowfall and, to a lesser extent, with wind speed, whereas the temperature estimate is small and imprecise. The subgroup estimates suggest that reductions in total hours are concentrated primarily among occupations with lower telework intensity. In the first quartile, colder temperatures and snowfall are associated with lower actual

working hours, and snowfall is also negatively associated with actual hours in the second quartile. By contrast, the estimates for the fourth quartile and the top decile are small relative to their standard errors, suggesting that total working hours in occupations with greater scope for remote work are less responsive to adverse winter conditions.

Panel B shows a different pattern for on-site work. In the full sample, lower temperatures, higher wind speed, and greater snowfall are all associated with reductions in on-site hours. The temperature estimates are similar in magnitude across occupational groups, though not always precisely estimated. The snowfall estimates, however, show a clearer occupational gradient. Snowfall reduces on-site hours across all occupational groups, and the magnitude of this reduction is generally larger among occupations with higher telework intensity, with the largest estimate in the top decile. The contrast with Panel A is informative: in the most telework-intensive occupations, snowfall sharply reduces on-site hours yet leaves total working hours largely unchanged. These patterns suggest that snowfall reduces on-site work across a wide range of occupations, but workers in occupations that are more telework-feasible appear better able to offset these reductions through additional remote work.

Figure 1 visualizes the *substitution effect* – where winter weather shifts working hours from on-site to remote work – and the *disruption effect*, whereby winter weather reduces both forms of work, decomposing the response of total working hours into its telework and on-site components across occupational telework-feasibility groups. In each panel, the blue bars report the change in telework hours, the red bars the change in on-site hours, and the solid line the net effect, defined as their sum and equal to the change in total actual working hours. The two effects can be read directly from the figure: when the net effect lies close to zero, the reduction in on-site hours is largely absorbed by an increase in telework, so that the substitution effect dominates; when the net effect is distinctly negative, winter weather lowers both on-site and telework hours, so that the disruption effect prevails.

Table 4. The impact of winter weather on actual and on-site working hours

	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: Dep. var. = Actual working hours						
Temperature	0.008	0.022**	0.006	0.011	-0.002	0.000
	(0.005)	(0.010)	(0.010)	(0.007)	(0.007)	(0.010)
Wind speed	-0.038*	-0.034	-0.063*	-0.061*	0.018	0.050
	(0.021)	(0.039)	(0.035)	(0.034)	(0.028)	(0.045)
Snowfall	-0.256***	-0.445*	-0.281*	-0.168	-0.159	-0.219
	(0.088)	(0.242)	(0.146)	(0.132)	(0.132)	(0.247)
Panel B: Dep. var. = On-site working hours						
Temperature	0.019***	0.021**	0.016	0.021**	0.018*	0.023
	(0.005)	(0.011)	(0.010)	(0.008)	(0.011)	(0.016)
Wind speed	-0.062***	-0.034	-0.069*	-0.083**	-0.035	-0.056
	(0.022)	(0.040)	(0.038)	(0.038)	(0.032)	(0.050)
Snowfall	-0.503***	-0.414*	-0.309**	-0.664***	-0.631***	-1.209***
	(0.073)	(0.229)	(0.141)	(0.166)	(0.187)	(0.260)
Observations	467,103	109,475	90,939	116,350	128,127	45,777

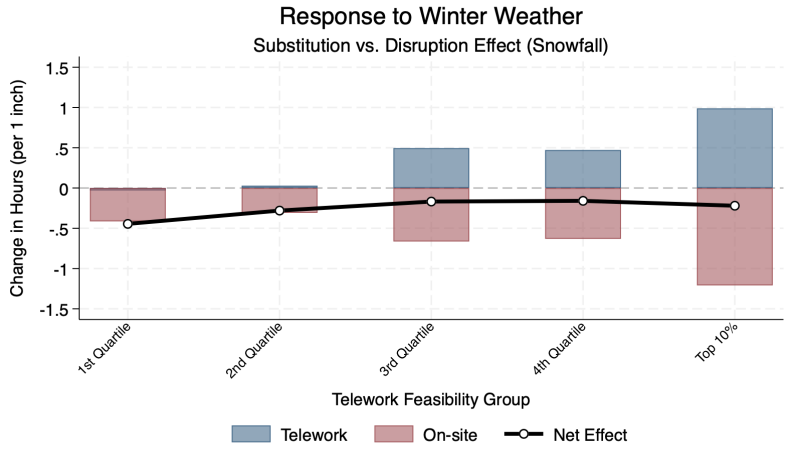
*** p<0.01, ** p<0.05, * p<0.1

Notes: This table reports coefficients from regressions of working hours on winter weather measures. The estimation sample includes workers in 193 metropolitan areas. Panel A uses total hours worked during the CPS reference week as the dependent variable. Panel B uses on-site working hours, defined as total hours worked minus telework hours. The weather variables are measured over the reference week and include average temperature, wind speed, and snowfall, reported in degrees Fahrenheit, miles per hour, and inches, respectively. Column (1) uses the full estimation sample, while Columns (2)–(5) restrict the sample to occupation-based quartiles defined by average telework hours. Column (6) restricts the sample to occupations in the top decile of average telework hours. All specifications include individual and year-month fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

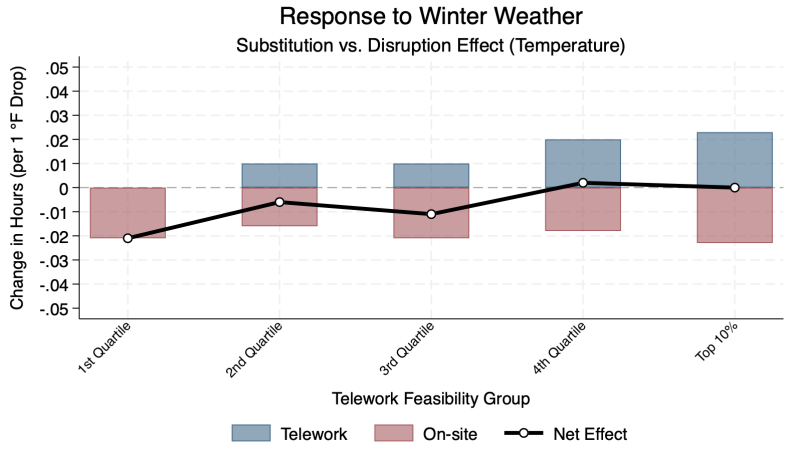
Panel (a) traces this decomposition across telework-feasibility groups for a one-inch increase in snowfall. In the lowest-feasibility group, snowfall reduces on-site hours by about 0.41 hours with essentially no telework response, so that the net decline of roughly 0.45 hours is almost entirely a disruption effect. As feasibility rises, both margins grow but increasingly offset one another: in the most telework-intensive occupations, an on-site reduction of about 1.21 hours is met by a telework increase of about 0.99 hours, shrinking the net decline to roughly 0.22 hours. The net effect line accordingly lies well below zero in the lowest-feasibility group and is substantially attenuated, though not eliminated, among the higher-feasibility groups, indicating that the substitution effect becomes more dominant where remote work is more feasible. Panel (b) shows the same qualitative gradient for a one-degree decrease in temperature, though the per-degree magnitudes are

far smaller and less precisely estimated, so the temperature decomposition corroborates rather than independently confirms the snowfall results. Visualized substitution and disruption effects of wind speed are provided in the Appendix.

Figure 1. Substitution vs. Disruption Effect of Winter Weather on Working Hours



(a) Snowfall



(b) Temperature

Notes: This figure visualizes the estimated effects of winter weather on telework hours and on-site working hours across occupation-based telework groups, using the regression estimates reported in Tables 2 and 4. Panel (a) shows the response to a one-inch increase in snowfall, and Panel (b) shows the response to a one-degree Fahrenheit decrease in temperature. Bars report the estimated changes in telework and on-site hours. The solid line with markers reports the net effect, defined as the sum of the telework and on-site hour changes, which corresponds to the estimated change in total working hours. Values closer to zero indicate that changes in telework hours offset changes in on-site hours more fully.

5.3 Dose-Response Estimates

Figure 2 presents the estimated dose-response relationship between snowfall intensity and telework hours from the continuous-treatment event-study framework of Callaway et al. (2024). The horizontal axis measures average snowfall at a worker's first month of snow exposure, while the vertical axis reports the estimated effect on telework hours. The solid red line summarizes the dose-response function estimated using a cubic spline with internal knots at the 25th, 50th, and 75th percentiles of the snowfall distribution among treated workers who experienced positive snowfall at their first-snow month[8], and the shaded area represents the 95 percent confidence interval. I interpret this curve as a smoothed summary of the estimated relationship over the observed support of snowfall, rather than as evidence tied to a particular parametric form.

The estimated curve is broadly consistent with the baseline results in Table 2. The point estimates rise with snowfall intensity over much of the observed range, indicating that workers exposed to heavier snowfall increase telework hours relative to those facing lower snowfall exposure. The confidence band widens at higher levels of snowfall, where observations are relatively sparse, so the upper-tail estimates should be read cautiously; nevertheless, the lower bound of the interval remains close to zero rather than turning negative, reinforcing the positive relationship implied by the estimates of the main specification. As a robustness check, the appendix reports a linear-spline version of the curve, which imposes fewer functional-form assumptions on the dose-response shape.

The dose-response estimates discussed so far pool all workers. Figure 3 instead estimates the relationship separately by occupational group, to examine whether the response to snowfall differs across occupations with different levels of telework feasibility. In general, the patterns are broadly consistent with the occupational heterogeneity documented in Table 2, though the curves in Figure 3 do not display a strictly monotonic

[8]Since snowfall is zero in many observations(see Table A1), placing the knots over the positive-snowfall distribution ensures that they fall within the range of snowfall intensities over which the dose-response is identified, rather than clustering at zero.

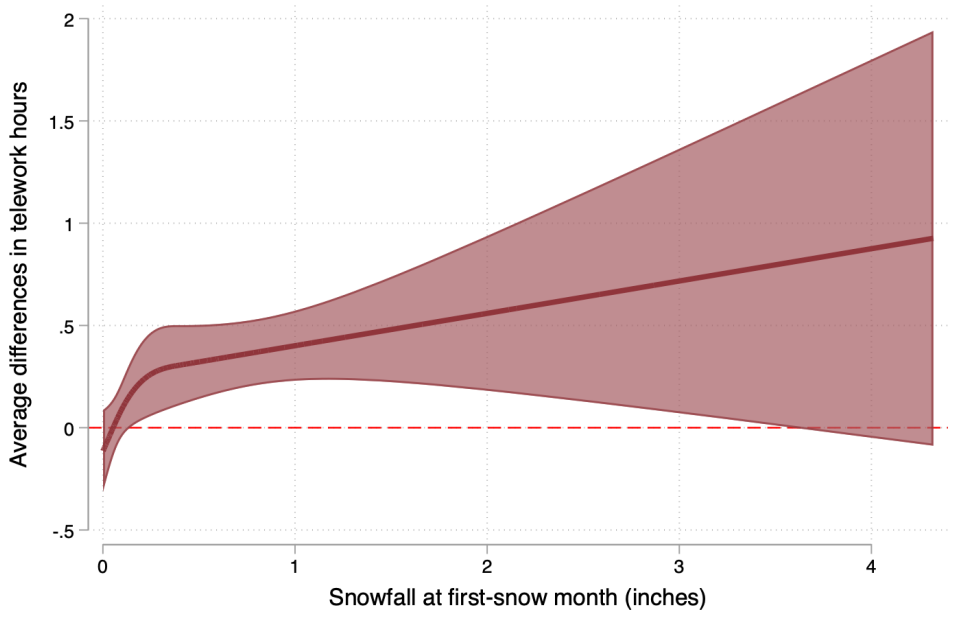
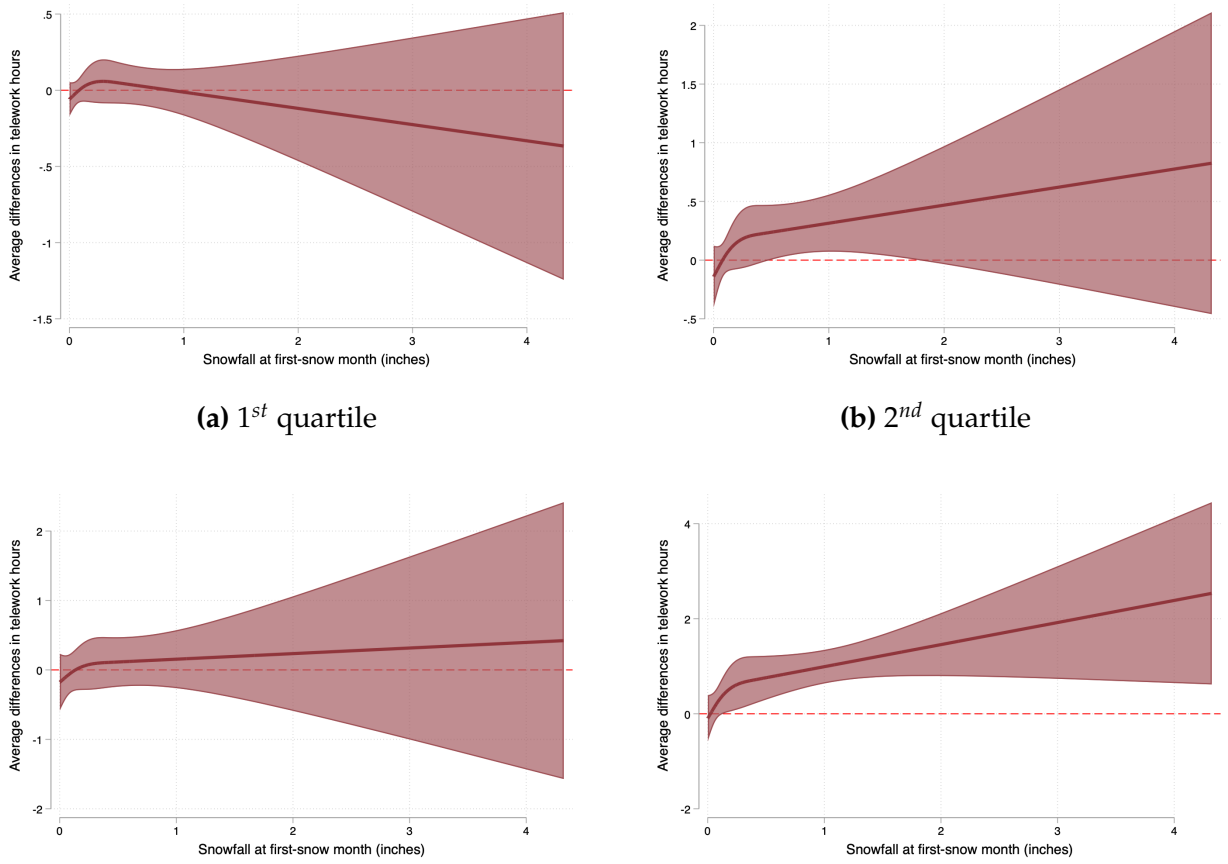


Figure 2. Estimated dose-response curve for the effects of snowfall

Notes: The figure shows the estimated dose-response function of telework hours with respect to snowfall intensity. Solid line denotes the estimate of $ATT^{es}_{e_1,e_2}(d)$ using Callaway, Goodman-Bacon, and Sant'Anna (2024). The shaded area is the 95% pointwise confidence intervals. The red line sets e_1 to 0 and e_2 to 1.

gradient across all quartiles. For occupations in the lowest quartile, the estimated response stays close to zero and turns slightly negative at higher snowfall levels, but the wide confidence interval makes this difference indistinguishable from zero. The second and third quartiles show more positive point estimates with similarly wide bands. The clearest pattern appears in the fourth quartile, where the curve rises more steadily over the observed range of snowfall. Consequently, these results reinforce the interpretation that telework responses to snowfall are concentrated among occupations with greater telework feasibility, while indicating that the gradient is more evident when comparing low- and high-telework-intensity occupations than when comparing adjacent quartiles.

Figure 3. Dose–response effects of snowfall on telework by occupational telework intensity



(a) 1st quartile

(b) 2nd quartile

(c) 3rd quartile

(d) 4th quartile

Notes: Each figure from (a) to (d) reports the estimated dose–response function between snowfall and telework hours for occupational groups with different levels of telework feasibility. Solid lines denote the estimate of $ATT^{es}_{e_1,e_2}(d)$ using Callaway, Goodman-Bacon, and Sant'Anna (2024). The shaded areas are the 95% pointwise confidence intervals. The red lines set e_1 to 0 and e_2 to 1.

5.4 Alternative Dimensions of Heterogeneity

While occupation-based categorization offers valuable insights, alternative classifications may also yield statistically significant results and implications. One such approach is regional categorization, as geography is inherently linked to climate and weather conditions. Table 5 examines whether the weather-induced changes in working arrangements differ across regions. I divide the sample into the nine Census divisions and estimate the baseline specification separately for each division. Because winter weather exposure,

transportation infrastructure, and occupational composition vary substantially across regions, these estimates provide a complementary way to assess whether the telework response documented above reflects a broader geographic adaptation pattern.

Panel A shows that the telework response to snowfall is concentrated in a limited set of divisions. The snowfall coefficients are positive in several northern divisions, with the most precisely estimated effects appearing in the East North Central and West North Central divisions and a weaker positive estimate in the Middle Atlantic. Temperature is statistically significant in more divisions than snowfall, including the Middle Atlantic, East North Central, West North Central, South Atlantic, and Pacific divisions, indicating that colder conditions are associated with higher telework hours across a broader set of regions. The pattern is nonetheless far from uniform: the snowfall estimates are small, imprecise, or of the opposite sign in several southern and western divisions, and wind speed is more relevant in parts of the West and West South Central regions. These results suggest that geography matters, but the telework response to winter weather is organized less systematically by region than by occupational telework intensity.

Panels B and C further illustrate this point. In Panel B, snowfall is associated with lower actual working hours in some divisions, including the South Atlantic, Mountain, and Pacific divisions, while the evidence is weaker or less consistent elsewhere. In Panel C, snowfall generally reduces on-site hours in several divisions, including the East North Central, West North Central, South Atlantic, Mountain, and Pacific divisions, with the notable exception of the West South Central division, where the estimate is positive and likely reflects the rarity of snowfall in that region. Crucially, these on-site reductions do not always coincide with positive telework responses in the same regions: in the South Atlantic and Pacific divisions, for example, on-site hours fall while the telework response is statistically insignificant. Adverse winter weather can therefore disrupt on-site labor across geographically diverse areas, but the extent to which workers offset these disruptions through telework varies across regions and does not follow a simple geographic

gradient. Relative to the occupational results in Tables 2 and 3, the regional estimates provide weaker evidence of the systematic substitution margin documented above.

Table 5. The impact of winter weather on working hours across divisions

	New England (1)	Middle Atlantic (2)	East North Central (3)	West North Central (4)	South Atlantic (5)	East South Central (6)	West South Central (7)	Mountain (8)	Pacific (9)
Panel A: Dep. var. = Telework hours									
Temperature	0.009	−0.056**	−0.042*	−0.058***	−0.062***	−0.030	−0.002	−0.010	−0.024**
	(0.044)	(0.026)	(0.024)	(0.018)	(0.015)	(0.017)	(0.009)	(0.011)	(0.011)
Wind speed	0.026	−0.001	0.022	0.026	−0.032	−0.050	0.049*	0.096*	0.062***
	(0.057)	(0.053)	(0.039)	(0.037)	(0.034)	(0.064)	(0.027)	(0.051)	(0.019)
Snowfall	0.303	0.330*	0.422***	0.280**	−0.637	0.003	−1.418	−0.041	0.716
	(0.224)	(0.180)	(0.151)	(0.119)	(0.417)	(1.964)	(0.953)	(0.091)	(0.667)
Panel B: Dep. var. = Actual working hours									
Temperature	0.038	−0.046	−0.054*	0.014	0.012	−0.022	0.025**	0.034	−0.015
	(0.061)	(0.036)	(0.028)	(0.054)	(0.021)	(0.025)	(0.011)	(0.020)	(0.022)
Wind speed	−0.034	−0.011	−0.071	−0.040	−0.169	−0.055	0.023	−0.067	−0.014
	(0.101)	(0.101)	(0.068)	(0.031)	(0.104)	(0.094)	(0.027)	(0.039)	(0.027)
Snowfall	−0.140	0.328	−0.246	−0.411	−2.590**	−2.073	0.386	−0.569***	−2.424***
	(0.368)	(0.204)	(0.180)	(0.242)	(1.207)	(2.935)	(0.661)	(0.099)	(0.812)
Panel C: Dep. var. = On-site working hours									
Temperature	0.029	0.010	−0.012	0.072	0.075***	0.008	0.028**	0.044*	0.008
	(0.072)	(0.045)	(0.032)	(0.063)	(0.023)	(0.026)	(0.013)	(0.021)	(0.027)
Wind speed	−0.060	−0.010	−0.093	−0.066	−0.138	−0.005	−0.026	−0.163***	−0.076**
	(0.104)	(0.104)	(0.061)	(0.048)	(0.110)	(0.114)	(0.045)	(0.048)	(0.035)
Snowfall	−0.443	−0.002	−0.669***	−0.692***	−1.954**	−2.075	1.804*	−0.528***	−3.140**
	(0.335)	(0.202)	(0.145)	(0.210)	(0.897)	(1.866)	(0.885)	(0.146)	(1.399)
Observations	44,538	18,834	61,055	35,777	78,366	25,629	69,460	49,045	84,399

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on telework hours (Panel A), actual working hours (Panel B), and on-site working hours (Panel C) among workers in 193 metropolitan areas, using a two-way fixed effects model. On-site working hours are defined as the difference between actual working hours and telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snowfall (inches). Each column presents results for one of the nine U.S. Census Bureau-designated divisions. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

Table 6 presents estimation results by educational attainment, motivated by the hypothesis that more educated workers hold jobs that are more amenable to telework and should therefore respond more strongly to winter weather. The estimates provide only partial support for this hypothesis. Among workers with a Bachelor's degree (Column 4), colder temperatures and heavier snowfall are both associated with significantly higher telework hours, consistent with the baseline results. For workers with a graduate degree (Column 5), however, neither the temperature nor the snowfall coefficient is statistically

significant, even though this is the group with the highest educational attainment. The absence of a consistent gradient across education levels indicates that educational attainment is not a robust criterion for organizing the telework response to winter weather. Given the strong correlation between education and occupational choice, the significant estimates in Table 6 are likely driven not by education itself, but by the concentration of telework-compatible occupations among more highly educated workers.

Table 6. The impact of winter weather on telework hours across education levels

	Telework hours				
	Full Sample	No HS Diploma	HS Graduate	Bachelor's Degree	Graduate Degree
	(1)	(2)	(3)	(4)	(5)
Temperature	-0.011***	0.003	-0.004	-0.020***	-0.013
	(0.003)	(0.006)	(0.003)	(0.006)	(0.009)
Wind speed	0.024**	-0.013	-0.004	0.035*	0.088***
	(0.011)	(0.015)	(0.012)	(0.020)	(0.033)
Snowfall	0.246***	-0.005	0.160**	0.365***	0.292
	(0.070)	(0.071)	(0.068)	(0.130)	(0.215)
Observations	467,103	32,310	180,228	172,395	75,214

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on telework hours among workers in 193 metropolitan areas, using a two-way fixed effects model. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snowfall (inches). Column (1) shows results for the full sample, while Columns (2) through (5) present estimates for groups with different levels of education. All models include individual and time fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

Age may also shape how workers respond to winter weather, for two reasons. First, telework-intensive occupations are likely distributed unevenly across age groups: computer-based and information-intensive jobs tend to be concentrated among prime-age and experienced workers who are more likely to hold positions with flexible work arrangements. Second, older individuals are generally more susceptible to cold-related health risks. Table 7 reports results by age group. The temperature coefficients are statistically significant for workers aged 45 and over but not for younger groups, suggesting that colder conditions raise telework hours primarily among older workers. This pattern is consistent with the first channel, under which telework-compatible occupations are concentrated among older workers, but it could also reflect the second channel, under which older individuals are more sensitive to cold weather. The two interpretations cannot be separated here, so

it is inconclusive whether age alone is a main driver of heterogeneous telework decisions.

The snowfall coefficient, by contrast, is positive and statistically significant only for the youngest groups, aged 15 to 24 (0.269) and 25 to 34 (0.487), and is insignificant among older workers. This is counterintuitive if older workers are more vulnerable to snowy outdoor conditions, and, together with the temperature results, it suggests that the health-vulnerability channel does not consistently explain the age pattern. One possibility is that younger workers are more exposed to the commuting disruptions caused by snowfall, for instance, through longer commutes or greater reliance on transportation that is more vulnerable to snow. As with education, the heterogeneity across age groups is weaker and less consistent than across occupational groups, so I treat this dimension as suggestive rather than central.

Table 7. The impact of winter weather on telework hours across age groups

	Telework hours						
	Full Sample	15–24	25–34	35–44	45–54	55–64	65+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Temperature	-0.011***	-0.000	-0.007	-0.009	-0.017**	-0.012*	-0.023**
	(0.003)	(0.005)	(0.007)	(0.006)	(0.008)	(0.007)	(0.011)
Wind speed	0.024**	0.024	0.043**	0.028	0.011	0.037	0.010
	(0.011)	(0.019)	(0.022)	(0.024)	(0.029)	(0.028)	(0.033)
Snowfall	0.246***	0.269**	0.487***	0.251	-0.023	0.228	0.236
	(0.070)	(0.123)	(0.185)	(0.154)	(0.130)	(0.181)	(0.238)
Observations	467,103	51,227	95,994	103,565	94,014	79,069	36,083

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents the estimated effects of winter weather on telework hours among workers in 193 metropolitan areas, using a two-way fixed effects model. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snowfall (inches). Column (1) shows results from the full sample, while Columns (2) through (7) present estimates for different age groups. All models include individual and time fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

5.5 Extensive-Margin Responses in Telework

The baseline results show that snowfall increases the number of hours worked from home, particularly among workers in occupations with greater baseline telework inten-

sity. This increase in telework hours may reflect two related forms of adjustment. Workers who would otherwise report no telework may begin working from home when winter weather conditions deteriorate, generating an extensive-margin response. Alternatively, workers who already work remotely may increase the number of hours they spend teleworking, generating an intensive-margin response. To examine the first channel, I estimate linear probability models in which the dependent variable is an indicator equal to one if a worker reports any positive telework hours during the CPS reference week.

Table 8 reports the results. The specification follows the baseline model, including individual and year-month fixed effects, the same weather variables, and standard errors clustered at the MSA level. The estimates show that snowfall increases the incidence of telework. In the full sample, an additional inch of snowfall raises the probability of reporting any telework by approximately 1.1 percentage points. This response is not uniform across occupations. The estimated effect is close to zero and statistically insignificant in the lowest quartile of occupational telework intensity, while it is larger and statistically significant among workers in the upper quartiles. In the top decile of occupations by baseline telework intensity, an additional inch of snowfall raises the probability of reporting any telework by about 3.0–3.2 percentage points.

These patterns suggest that part of the increase in telework hours documented in the baseline results operates through the extensive margin. Winter weather appears to induce some workers, especially those in occupations where remote work is more feasible, to report positive telework hours during the reference week. At the same time, the extensive-margin estimates should be interpreted as changes in realized telework incidence rather than as direct observations of workers' unconstrained telework choices, since the ability to telework may also depend on job tasks, employer policies, and workplace arrangements.

Table 8. The impact of winter weather on the extensive margin of telework across occupational groups

	Any telework experience in a reference week					
	Full Sample	1st Quartile	2nd Quartile	3rd Quartile	4th Quartile	Top 10 percent
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Weekly average temperature						
Temperature	-0.000**	-0.000	-0.000*	-0.000	-0.001*	-0.001*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Wind speed	0.001**	0.000	0.000	0.001	0.002**	0.003**
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
Snowfall	0.011***	-0.001	0.005	0.019***	0.016***	0.032***
	(0.002)	(0.002)	(0.003)	(0.005)	(0.006)	(0.009)
Panel B: Temperature bin						
Low-temperature bin						
(20°F or below)	-0.000	-0.000	0.001	0.002	-0.001	0.004
	(0.001)	(0.001)	(0.002)	(0.002)	(0.003)	(0.003)
Mid-temperature bin						
(20°F to 40°F)	0.001*	-0.000	0.001	0.001	0.003	0.004
	(0.001)	(0.000)	(0.001)	(0.001)	(0.002)	(0.003)
High-temperature bin						
(40°F to 60°F)	0.000	-0.000	-0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
Wind speed	0.001**	0.000	0.000	0.001	0.002**	0.003**
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
Snowfall	0.011***	-0.001	0.004	0.019***	0.015**	0.030***
	(0.002)	(0.002)	(0.003)	(0.005)	(0.006)	(0.009)
Observations	467,103	109,475	90,939	116,350	128,127	45,777

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table reports coefficients from linear probability models of telework incidence on winter weather measures. The dependent variable is an indicator equal to one if a worker reports any positive telework hours during the CPS reference week, and zero otherwise. Coefficients can be interpreted as changes in the probability of reporting any telework; multiplying the coefficients by 100 gives percentage-point changes. The full estimation sample includes workers in 193 metropolitan areas. Panel A uses continuous measures of average temperature, wind speed, and snowfall, reported in degrees Fahrenheit, miles per hour, and inches, respectively. Panel B replaces the continuous temperature measure with temperature-bin indicators, with 60°F or above as the omitted category. Column (1) uses the full estimation sample, while Columns (2)–(5) restrict the sample to occupation-based quartiles defined by average telework hours. Column (6) restricts the sample to occupations in the top decile of average telework hours. All specifications include individual and year-month fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

6 Robustness Checks

Table 9 replicates the baseline specification, replacing snowfall with snow depth as the measure of winter precipitation. As discussed in Section 3.1, the two variables capture distinct dimensions of snow exposure: daily snowfall reflects newly fallen snow during the last 24 hours since the last measurement and thus represents the immediate disruption of an ongoing snow event, whereas snow depth reflects accumulated snowpack on

the ground and the more persistent mobility frictions associated with it. Re-estimating the baseline model with snow depth therefore tests whether the telework response is specific to acute snowfall shocks or extends to persistent snow conditions. Because snow depth observation is missing for some station-weeks, the sample size is smaller than in the baseline.

The results are broadly consistent with the baseline findings. In Panel A, snow depth is positive and statistically significant for the full sample, and, as with snowfall, the effect is concentrated among occupations with higher telework feasibility: the estimates are insignificant in the 1st and 2nd quartiles but rise up to 0.189 in the top decile. Panel B, which replaces the continuous temperature term with bins, preserves the same snow-related pattern: snow depth remains positive, significant, and increasing in telework feasibility. Meanwhile, the temperature-bin estimates themselves differ somewhat from those in Table 2, with several bins becoming significant in the full sample and in the middle quartiles. The coefficients on snow depth are not directly comparable in magnitude to the baseline snowfall estimates, since the two measures differ in scale and distribution; what matters for robustness is that the sign, significance, and occupational gradient of the snow effect are preserved. Overall, the positive relationship between winter precipitation and telework is not sensitive to how snow exposure is measured, supporting the interpretation that snow-related disruptions prompt a reallocation of work from on-site to remote settings.

A potential concern with the occupation-based heterogeneity analysis is that some workers change occupations during the CPS sample period. Such transitions may mechanically change a worker's assigned telework-feasibility group, making it difficult to distinguish weather-induced telework responses from changes in occupational composition. In addition, although likely less common over the short horizon considered here, severe winter conditions could discourage some workers from leaving or switching out of occupations that are especially disrupted by cold or snow. To address these concerns,

Table 9. Robustness Check: Replacing Snowfall with Snow depth

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Panel A: Weekly average temperature						
Temperature	-0.011***	0.002	-0.012***	-0.014**	-0.017	-0.024
	(0.004)	(0.002)	(0.004)	(0.006)	(0.011)	(0.018)
Wind speed	0.019*	0.002	0.002	0.024	0.049	0.116**
	(0.011)	(0.006)	(0.012)	(0.021)	(0.033)	(0.051)
Snow depth	0.036***	-0.004	-0.003	0.065**	0.087**	0.189***
	(0.013)	(0.010)	(0.014)	(0.032)	(0.034)	(0.064)
Panel B: Temperature bin						
Low-temperature bin (20°F or below)	0.057*	-0.012	0.086**	0.158**	0.017	0.145
	(0.031)	(0.021)	(0.043)	(0.063)	(0.086)	(0.153)
Mid-temperature bin (20°F to 40°F)	0.059***	-0.022*	0.046**	0.090**	0.087	0.100
	(0.022)	(0.012)	(0.020)	(0.039)	(0.058)	(0.100)
High-temperature bin (40°F to 60°F)	0.031**	-0.012	0.023*	0.062**	0.012	-0.021
	(0.014)	(0.010)	(0.013)	(0.031)	(0.038)	(0.061)
Wind speed	0.018	0.003	0.001	0.022	0.046	0.107**
	(0.012)	(0.006)	(0.012)	(0.021)	(0.033)	(0.052)
Snow depth	0.044***	-0.006	-0.003	0.065*	0.097***	0.165***
	(0.014)	(0.010)	(0.016)	(0.036)	(0.036)	(0.060)
Observations	391,718	91,639	77,332	97,571	106,479	38,088

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table replicates the baseline specification from Table 2, replacing snowfall with snow depth (inches) as a measure of winter precipitation. The dependent variable is the number of telework hours. In Panel A, the independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snow depth (inches). Panel B replaces the continuous temperature variable with three temperature bins. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

Table 10 restricts the sample to occupation stayers, defined as workers whose reported occupation remains unchanged during the sample period. This restriction holds occupational telework feasibility fixed within workers and therefore provides a cleaner check of whether the baseline heterogeneity patterns are driven by changes in work arrangements rather than by occupational switching.

The results remain broadly similar to the baseline estimates. Although the sample size falls substantially, the snowfall response remains concentrated among occupations

with greater telework feasibility, while the estimates for the lowest-telework occupations remain small. This pattern suggests that the main results are not primarily driven by workers moving across occupational groups during the survey period.

Table 10. Robustness Check: Restricting to Occupation Stayers

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Temperature	-0.009**	-0.002	-0.003	-0.009	-0.018	-0.021
	(0.004)	(0.003)	(0.004)	(0.007)	(0.012)	(0.021)
Wind speed	0.017	0.006	0.003	0.038	0.016	-0.001
	(0.014)	(0.008)	(0.013)	(0.025)	(0.039)	(0.063)
Snowfall	0.276***	-0.081	0.096	0.225	0.722***	0.940***
	(0.071)	(0.054)	(0.065)	(0.147)	(0.247)	(0.291)
Observations	265,741	57,931	63,233	66,866	77,711	24,363

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table replicates the baseline specification from Panel A of Table 2, but restricts the sample to workers who did not change occupations during the sample period. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snowfall (inches). Column (1) shows results from the full sample of occupation stayers, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

Table 11 reports a placebo test that adds one-month lead weather variables to the baseline specification. The purpose of this exercise is to assess whether future weather conditions predict current telework hours after controlling for contemporaneous weather. If the baseline estimates primarily reflect short-run responses to weather conditions during the CPS reference week, weather realized one month later should not be systematically related to current telework decisions.

The results are consistent with this expectation. The contemporaneous weather coefficients remain close to the baseline estimates, particularly for snowfall in the full sample and in occupations with higher telework intensity. By contrast, the lead weather variables are not precisely estimated in any column. The coefficients on lead temperature are close to zero across the occupational groups, and the lead snowfall estimates do not exhibit a systematic pattern. Lead wind speed is also imprecisely estimated throughout. These findings provide reassurance that the baseline results are unlikely to be driven by per-

sistent seasonal patterns or other unobserved factors that would cause future weather to predict current telework hours. Instead, the evidence is more consistent with a contemporaneous adjustment of telework hours to winter weather conditions.

Table 11. Placebo Test: Using Lead Weather Variables

	Telework hours					
	Full Sample	1st Quartile	2nd Quartile	3rd Quartile	4th Quartile	Top 10 percent
	(1)	(2)	(3)	(4)	(5)	(6)
Temperature	-0.011***	0.001	-0.011***	-0.010*	-0.020**	-0.021
	(0.003)	(0.002)	(0.004)	(0.006)	(0.010)	(0.015)
Wind speed	0.025**	0.000	0.007	0.027	0.051*	0.110**
	(0.011)	(0.007)	(0.012)	(0.020)	(0.029)	(0.046)
Snowfall	0.250***	-0.015	0.038	0.508***	0.469**	1.056***
	(0.076)	(0.044)	(0.057)	(0.131)	(0.218)	(0.320)
Temperature$_{t+1}$	0.001	0.001	-0.001	0.003	0.000	0.008
	(0.003)	(0.002)	(0.003)	(0.006)	(0.007)	(0.015)
Wind speed$_{t+1}$	0.012	-0.002	0.019	0.019	0.021	0.049
	(0.012)	(0.007)	(0.012)	(0.023)	(0.033)	(0.051)
Snowfall$_{t+1}$	-0.013	0.068	-0.016	-0.021	0.074	0.287
	(0.084)	(0.045)	(0.064)	(0.107)	(0.209)	(0.365)
Observations	465,896	109,148	90,657	116,071	127,853	45,708

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table presents results from a placebo test in which one-month lead weather variables are included alongside contemporaneous weather conditions, using the main specification. If the baseline results reflect a causal relationship between weather and telework, future weather conditions should have no predictive power for current telework hours after controlling for contemporaneous weather. The dependent variable is the number of telework hours. The independent variables include weekly averages of temperature (°F), wind speed (miles per hour), and snowfall (inches), measured both in the CPS reference week and one month after. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

7 Conclusion

This paper examines whether remote work serves as a short-run margin of adaptation to adverse winter weather. Combining individual-level telework data from the Current Population Survey with daily weather records from NOAA, I estimate how winter weather affects telework hours, total working hours, and on-site working hours across U.S. metropolitan areas. Winter weather increases telework hours, with the clearest and most consistent effect arising from snowfall. Greater wind speed also raises telework

hours, particularly in more telework-intensive occupations, whereas the temperature estimates are weaker and less stable. This response is concentrated among occupations with greater telework intensity, which I use as a proxy for occupation-specific telework feasibility.

The increase in telework reflects a broader reallocation of work across locations rather than a net rise in hours. Snowfall reduces on-site working hours across all occupational groups, but the accompanying decline in total working hours is much smaller among occupations with greater scope for remote work. As the decomposition in Figure 1 makes clear, the substitution effect dominates among telework-feasible occupations, where lost on-site hours are largely converted into remote hours, whereas the disruption effect prevails among occupations with little scope for telework, where both margins fall. Remote work, therefore, provides a real but partial adaptation margin, available primarily to workers whose tasks can be performed away from the workplace. The average magnitude of this adaptation is moderate, but this reflects the nature of the shock and the population rather than a weak underlying response. Most snowfall events are too light to meaningfully impede commuting, and the capacity to substitute toward remote work is concentrated among telework-feasible occupations; where snowfall is heavy enough to disrupt commuting and remote work is feasible, the response is substantial. As remote work continues to diffuse across occupations, the scope for this margin of adaptation may widen accordingly.

This paper proposes that occupational telework feasibility is the most systematic dimension along which weather responses differ. Regional, educational, and age-based classifications reveal some variation, but these patterns are less consistent than the occupational gradient and appear to operate largely through the occupational composition of each group. The continuous-treatment dose-response estimates reinforce the baseline findings, tracing a positive relationship between snowfall intensity and telework hours that holds across most of the observed range. Three robustness exercises further support

the baseline results: they are preserved under an alternative snow measure (snow depth), within a sample restricted to occupation stayers, and in a lead-weather placebo test in which future weather has no predictive power for current telework.

These findings shed light on how labor markets adapt to weather shocks. With remote work now more prevalent than before the COVID-19 pandemic, it can mitigate the reduction in hours worked caused by adverse winter weather for workers whose jobs are compatible with telework. These benefits, however, are not evenly distributed across the workforce. Policies and firm practices that expand remote-work capacity, digital infrastructure, and flexible arrangements may strengthen resilience to weather disruptions, but their reach is bounded by the set of occupations in which remote work is technically and organizationally feasible. More broadly, the results suggest that spatial reallocation of work is an important form of adaptation to winter weather.

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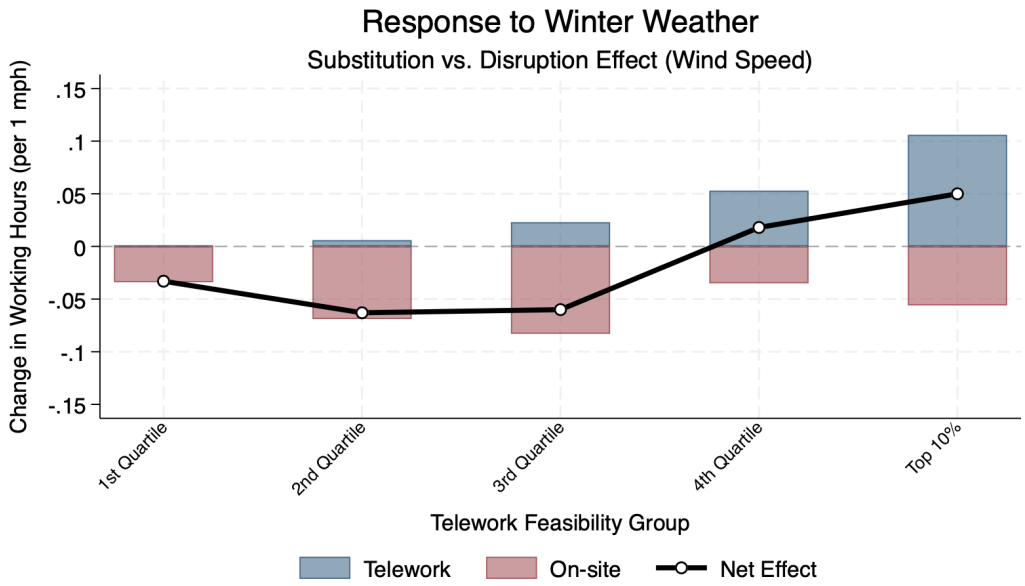
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Appendix

Figure A1. Substitution vs. Disruption Effect of Wind Speed on Working Hours



Notes: This figure visualizes the estimated effects of wind speed on telework hours and on-site working hours across occupation-based telework groups, using the regression estimates reported in Tables 2 and 4. The figure shows the response to a one-mile-per-hour increase in average wind speed. Bars report the estimated changes in telework and on-site hours: the blue bars report the change in telework hours and the red bars the change in on-site hours. The solid line with markers reports the net effect, defined as the sum of the telework and on-site hour changes, which corresponds to the estimated change in total working hours. Values closer to zero indicate that changes in telework hours offset changes in on-site hours more fully.

Table A1. Summary Statistics of Weather Variables

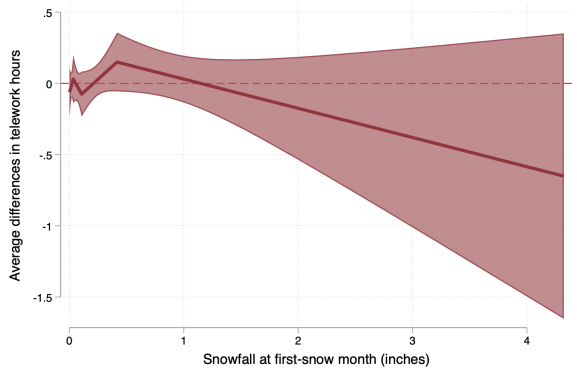
	Snowfall (inches)	Snow Depth (inches)
Panel A: Snow Measures		
Mean	0.08	0.38
SD	0.27	1.53
Share $= 0$ (%)	77.3	79.3
Conditional on > 0		
Mean	0.35	1.82
Median	0.14	0.64
P90	0.88	4.65
P99	2.29	14.16
Observations	467,103	391,718

	Temperature (°F)
Panel B: Temperature	
Mean	50.37
SD	14.12
Median	51.53
Exposure days per reference week	
$<$ 20°F	0.18
20–40°F	1.58
40–60°F	3.37
$>$ 60°F	1.87

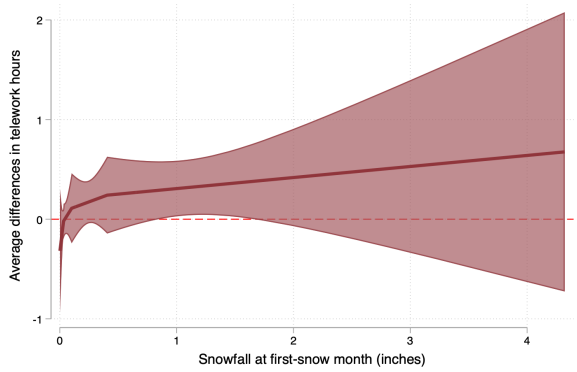
	Wind Speed (mph)
Panel C: Wind Speed	
Mean	7.87
SD	2.65
Median	7.77
Distribution	
P90	11.40
P99	14.86
Observations	467,103

Notes: The sample consists of CPS respondents observed during October–March, 2022–2025. Snowfall and snow depth are measured as average daily values over the seven-day CPS reference week. Temperature exposure days count the number of days during the reference week in which the average daily temperature falls within the indicated bin. Wind speed is measured as the average daily wind speed over the reference week.

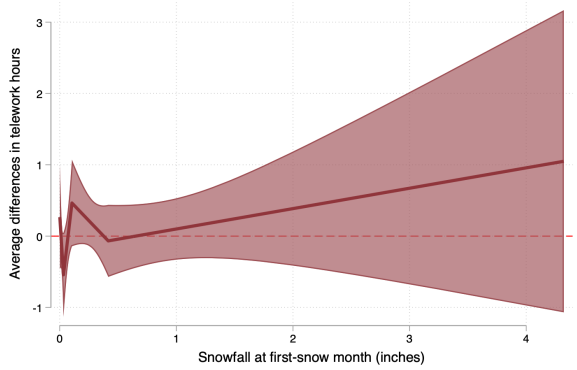
Figure A2. Dose–response effects of snowfall on telework by occupational telework intensity in linear spline



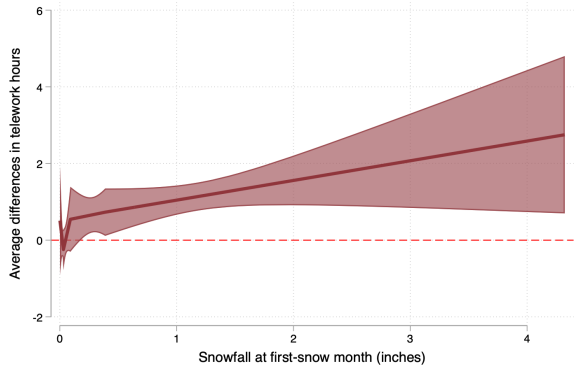
(a) 1st quartile



(b) 2nd quartile



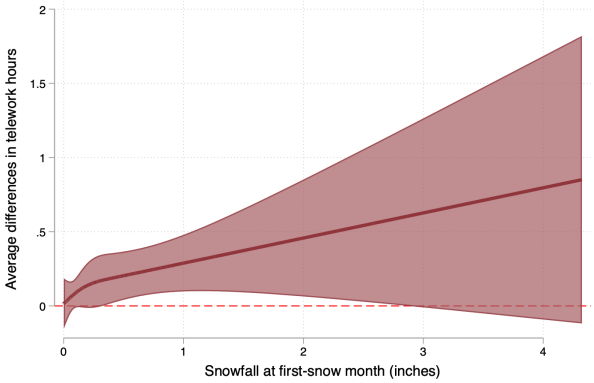
(c) 3rd quartile



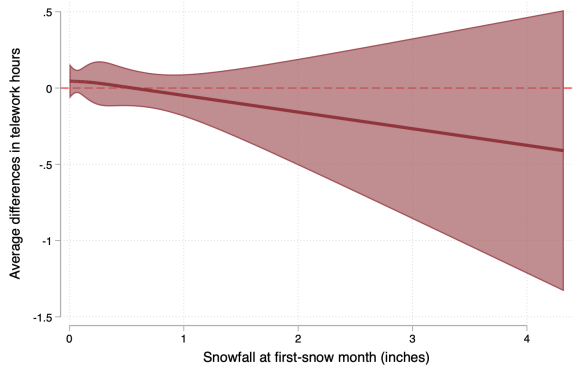
(d) 4th quartile

Notes: Each figure from (a) to (d) reports the estimated dose–response function between snowfall and telework hours for occupational groups with different levels of telework feasibility. Solid lines denote the estimate of $ATT^{es}_{e_1,e_2}(d)$ using Callaway, Goodman-Bacon, and Sant'Anna (2024). The shaded areas are the 95% pointwise confidence intervals. The red lines set e_1 to 0 and e_2 to 1.

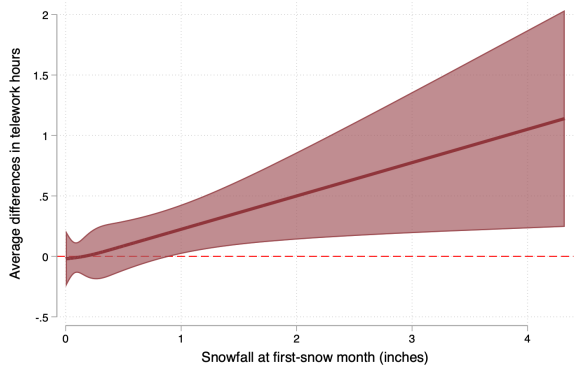
Figure A3. Dose–response effects of snowfall on telework with a smaller event window ($e_1 = e_2 = 0$)



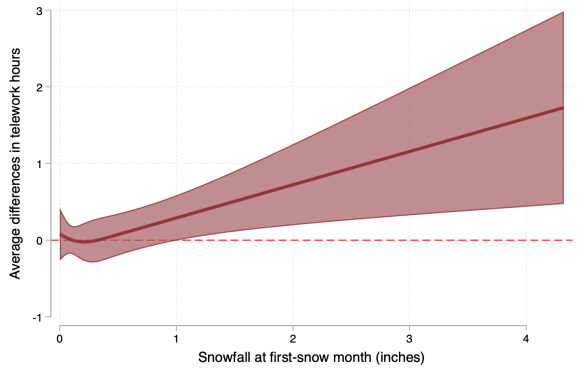
(a) Full sample



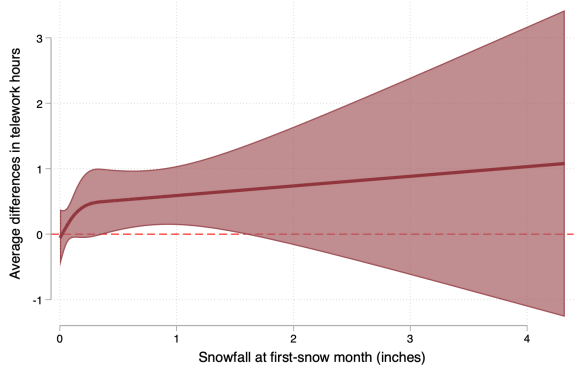
(b) 1st quartile



(c) 2nd quartile



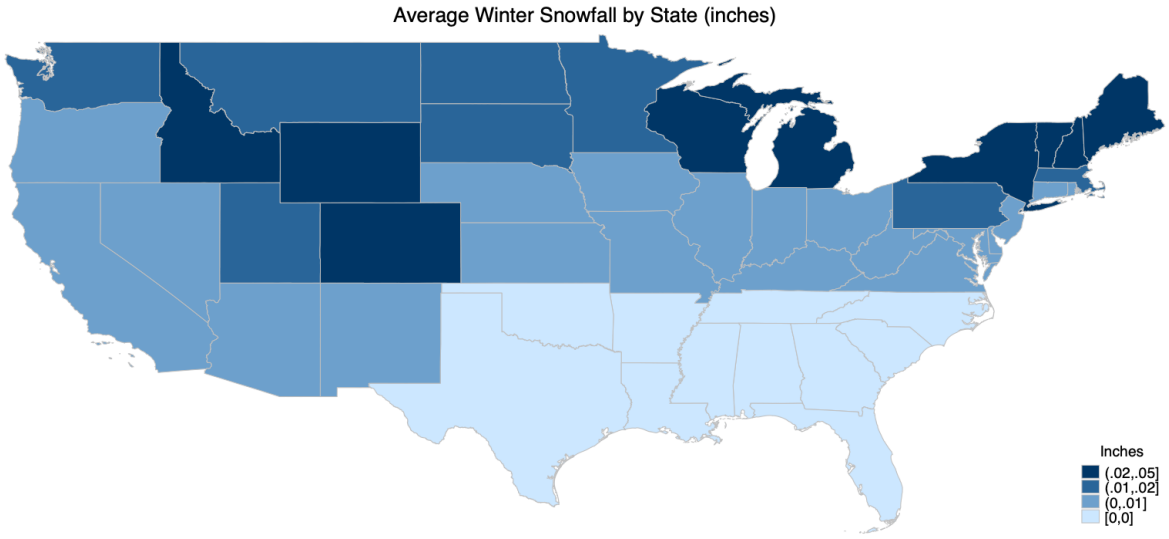
(d) 3rd quartile



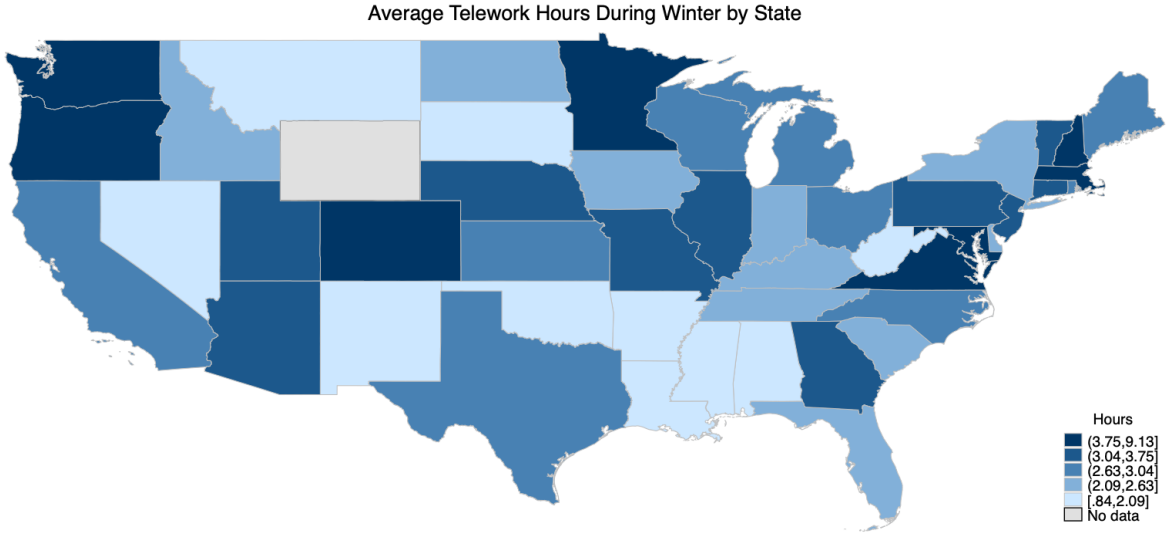
(e) 4th quartile

Notes: Panels (a)–(e) report the estimated dose–response function between snowfall and telework hours for the full sample and for occupational groups using the full sample (Panel (a)) or at different levels of telework feasibility(Panels (b) – (e)). Solid lines denote the estimate of $ATT^{es}_{e_1,e_2}(d)$ using Callaway, Goodman-Bacon, and Sant'Anna (2024). Shaded areas are 95% pointwise confidence intervals. The red lines set $e_1 = e_2 = 0$. By limiting the event window to the main identification, these dose-response curves estimate workers' immediate response to the snowfall shock.

Figure A4. Winter Snow Depth and Telework Hours by State



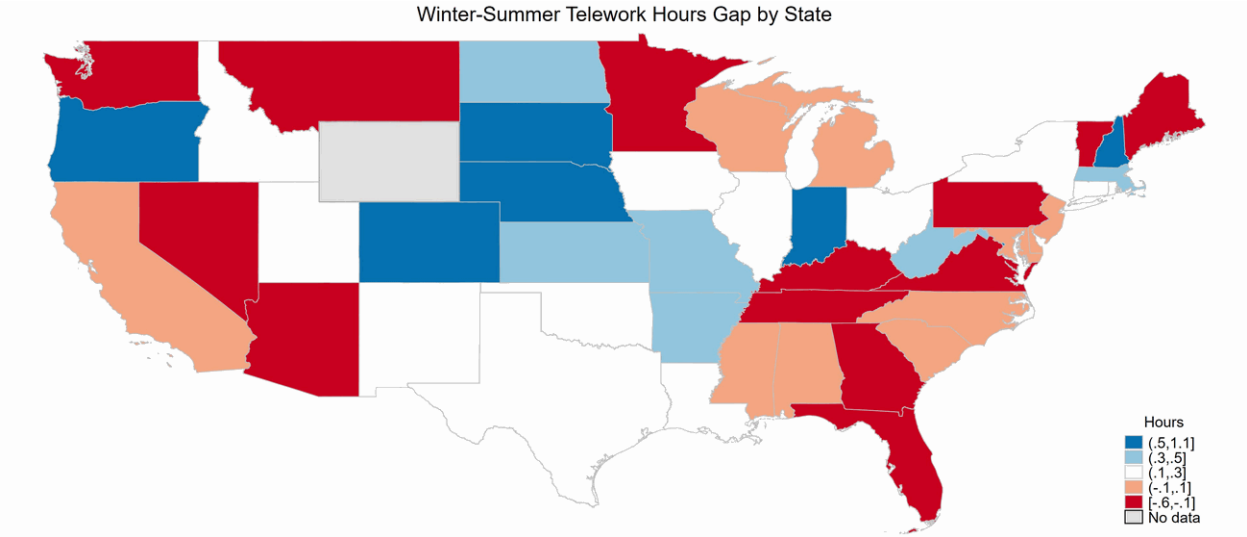
(a) Average winter snowfall



(b) Average winter telework hours

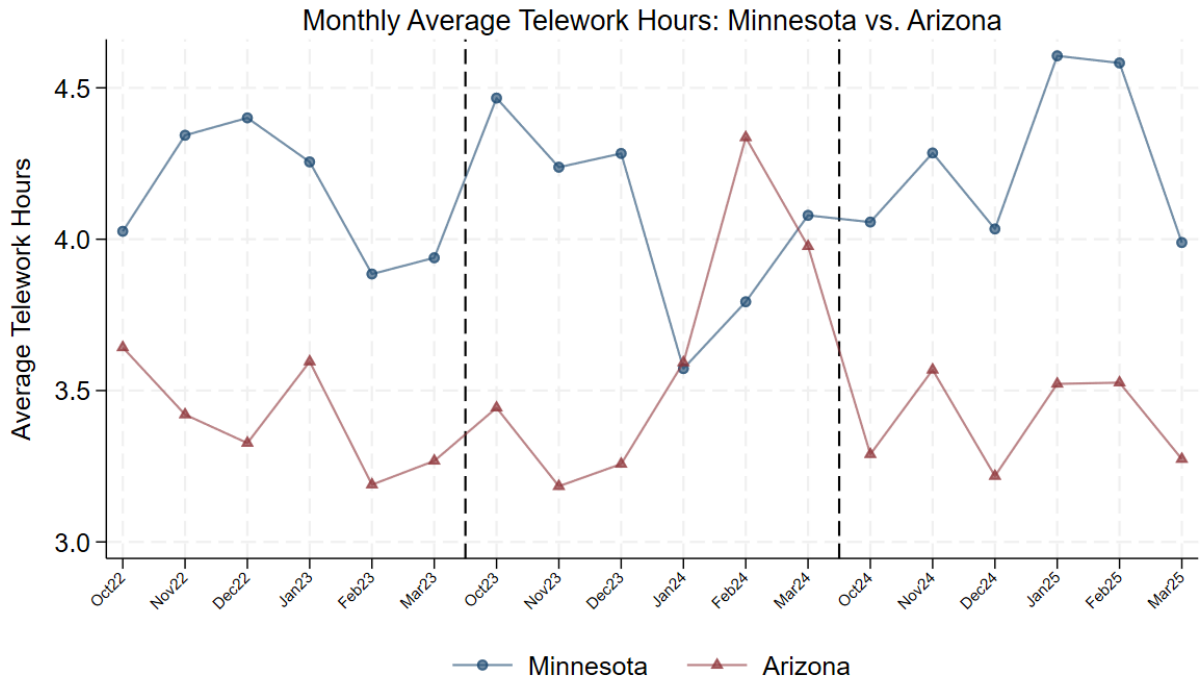
Notes: Panel (a) displays the average winter snowfall (in inches) by state, and Panel (b) displays the average winter telework hours by state. Both panels are based on the 2022–2025 sample period, with winter defined as October through March. Snowfall data are drawn from NOAA weather stations matched to 193 metropolitan statistical areas, and telework hours are drawn from the Current Population Survey. States shown in gray indicate missing data. Alaska and Hawaii are excluded.

Figure A5. Winter-Summer Telework Hours Gap by State



Notes: This figure displays the difference between average winter and summer telework hours by state. Winter is defined as December and January, and summer is defined as June and July. Blue shading indicates states where telework hours are higher in winter than in summer (positive gap), while red shading indicates states where telework hours are lower in winter (negative gap). The data cover the 2022–2025 sample period, drawn from the Current Population Survey. State-level averages are computed by aggregating individual-level telework hours across all workers within each state. States shown in gray indicate missing data. Alaska and Hawaii are excluded.

Figure A6. Monthly Average Telework Hours: Minnesota vs. Arizona



Notes: This figure compares the monthly average telework hours between Minnesota (a cold-climate state) and Arizona (a warm-climate state) over the October 2022 to March 2025 sample period. Only winter months (October through March) are shown, with summer months excluded. Vertical dashed lines indicate boundaries between winter seasons. Telework hours are drawn from the Current Population Survey and averaged across all workers within each state for a given month. Minnesota consistently exhibits higher telework hours than Arizona, with the gap widening during the coldest months (December through February), suggesting a positive relationship between cold weather and telework behavior.

Table A2. Robustness Check: Finer Temperature Bins

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Wind speed	0.023*	0.001	0.006	0.023	0.046	0.087*
	(0.012)	(0.006)	(0.012)	(0.020)	(0.030)	(0.046)
Snowfall	0.259***	-0.035	0.044	0.528***	0.417**	0.903***
	(0.074)	(0.049)	(0.057)	(0.117)	(0.211)	(0.273)
Days with mean temperature in range:						
$<$ 10°F	0.089***	0.015	0.198***	0.175*	-0.053	-0.277
	(0.034)	(0.029)	(0.056)	(0.092)	(0.126)	(0.237)
10°F to 20°F	0.027	-0.028	0.019	0.106*	0.047	0.356**
	(0.036)	(0.027)	(0.045)	(0.063)	(0.113)	(0.169)
20°F to 30°F	0.034	-0.009	0.029	-0.004	0.115*	0.028
	(0.024)	(0.012)	(0.026)	(0.042)	(0.063)	(0.116)
30°F to 40°F	0.040*	-0.023*	0.037*	0.058*	0.049	0.045
	(0.021)	(0.012)	(0.019)	(0.034)	(0.051)	(0.094)
40°F to 50°F	0.017	-0.014	0.022	0.033	-0.001	-0.005
	(0.015)	(0.012)	(0.017)	(0.029)	(0.040)	(0.072)
50°F to 60°F	0.005	-0.012*	0.007	0.017	-0.017	-0.080
	(0.014)	(0.007)	(0.012)	(0.032)	(0.032)	(0.053)
Observations	467,103	109,475	90,939	116,350	128,127	45,777

*** p$<$0.01, ** p$<$0.05, * p$<$0.1

Notes: This table replicates the temperature bin specification from Panel B of Table 2, but with finer bin widths of 10°F instead of 20°F. Each temperature-bin variable counts the number of days in the reference week with an average temperature within the specified range. The omitted category is days with temperatures above 60°F. The dependent variable is the number of telework hours. Column (1) shows results from the full sample, while Columns (2) through (5) present estimates for occupation-based quartile groups defined by average telework hours. Column (6) shows the results for the top decile of occupations with the highest average telework hours. The table reports estimated coefficients with robust standard errors in parentheses. All models include individual and time fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

Table A3. The impact of snowfall intensity on telework hours across occupational groups

	Telework hours					
	Full Sample (1)	1st Quartile (2)	2nd Quartile (3)	3rd Quartile (4)	4th Quartile (5)	Top 10 percent (6)
Light snowfall bin (0 to 0.5 inches)	0.084*	-0.002	0.112**	0.112	0.285**	0.630**
	(0.047)	(0.033)	(0.055)	(0.090)	(0.122)	(0.270)
Moderate snowfall bin (0.5 to 1.5 inches)	0.102	0.068	0.032	0.144	0.168	0.815*
	(0.091)	(0.054)	(0.089)	(0.189)	(0.205)	(0.442)
Heavy snowfall bin (above 1.5 inches)	0.488***	-0.159	0.275*	1.120***	0.881	1.417**
	(0.174)	(0.126)	(0.153)	(0.300)	(0.569)	(0.599)
Temperature	-0.011***	0.001	-0.009***	-0.012**	-0.020**	-0.021
	(0.003)	(0.002)	(0.003)	(0.005)	(0.010)	(0.016)
Wind speed	0.024**	0.001	0.005	0.023	0.053*	0.107**
	(0.011)	(0.006)	(0.012)	(0.020)	(0.030)	(0.045)
Observations	467,103	109,475	90,939	116,350	128,127	45,777

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table replicates the baseline specification from Panel A of Table 2, replacing the continuous snowfall measure with snowfall-bin indicators. Each bin indicates whether the average daily snowfall over the CPS reference week falls within the specified range, measured in inches; reference weeks with no snowfall constitute the omitted category. The dependent variable is the number of hours worked from home during the CPS reference week. Temperature and wind speed are weekly averages reported in degrees Fahrenheit and miles per hour, respectively. Column (1) uses the full estimation sample, while Columns (2)–(5) restrict the sample to occupation-based quartiles defined by average telework hours. Column (6) restricts the sample to occupations in the top decile of average telework hours. All specifications include individual and year-month fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

Table A4. Decomposition of the telework response to snowfall by telework status, top decile occupations

	Telework hours		Any telework
	Always	Switchers	Switchers
	(1)	(2)	(3)
Temperature	0.019	-0.093^{***}	-0.0026^{**}
	(0.016)	(0.034)	(0.0012)
Wind speed	0.063	0.154	0.0064^{**}
	(0.046)	(0.105)	(0.0029)
Snowfall	0.253	1.389^{***}	0.0585^{***}
	(0.258)	(0.477)	(0.0191)
Observations	18,559	17,000	17,000
Share of sample (%)	41.3	37.1	37.1

Notes: This table decomposes the telework response to snowfall among workers in occupations in the top decile of average telework hours. Workers are classified by their telework status across all observed CPS reference weeks: those reporting positive telework hours in every week (Always), those reporting positive hours in some weeks but not others (Switchers), and those never reporting positive hours. Workers in the last group account for 21.6 percent of the sample and are omitted from the table, since they contribute no within-person variation in either outcome and all estimated coefficients are zero for that group. Columns (1) and (2) report coefficients from regressions of telework hours on winter weather measures; Column (3) reports a linear probability model in which the dependent variable is an indicator for any positive telework hours. The ratio of the snowfall coefficients in Columns (2) and (3) implies average telework hours of 23.7 for workers induced into telework by snowfall. Shares are computed over person-weeks using the CPS final person weight. All specifications include individual and year-month fixed effects. Standard errors clustered at the MSA level are reported in parentheses.

Significance: *** $p<0.01$, ** $p<0.05$, * $p<0.1$